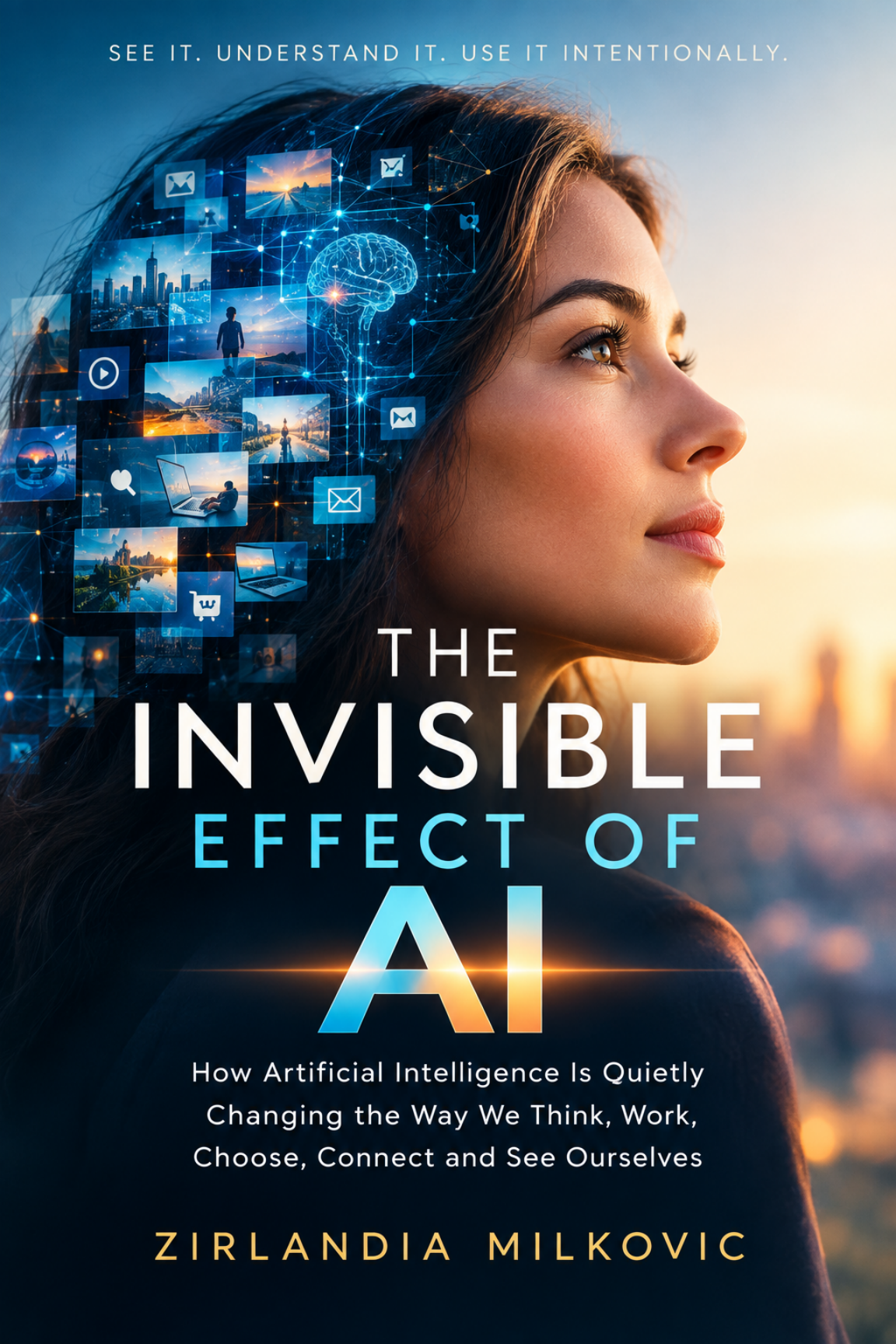
SEE IT. UNDERSTAND IT. USE IT INTENTIONALLY.



THE INVISIBLE EFFECT OF AI

How Artificial Intelligence Is Quietly Changing the Way We Think, Work, Choose, Connect and See Ourselves

ZIRLANDIA MILKOVIC

THE INVISIBLE EFFECT OF AI

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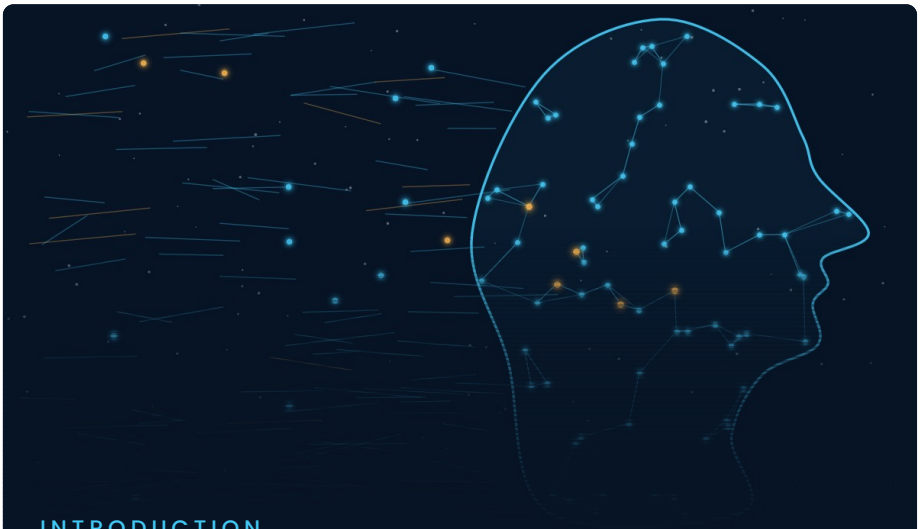


INTRODUCTION

The Technology You Don't Notice Anymore

Elena's alarm goes off at 6:40. She did not set it for 6:40. Her phone did, after noticing that on weekday mornings she tends to lie awake for a few minutes before getting up, and that on days with an early meeting she prefers a gentler tone. She doesn't know this. She simply thinks of it as "my alarm."

She unlocks the phone with her face. The lock screen shows a photo the phone selected from thousands, a weather summary written for her location, and three headlines chosen by a system that has been learning her reading habits for years. She skims them while the kettle boils. One story makes her frown. Another makes her smile. She will not remember, later, that neither was chosen by an editor who knew her.

On the commute, her navigation app reroutes her around an accident she never saw. A music service queues a playlist that feels uncannily right for a grey morning. Her email program finishes her sentences as she types them. Her spam filter quietly discards forty messages she will never know existed. At

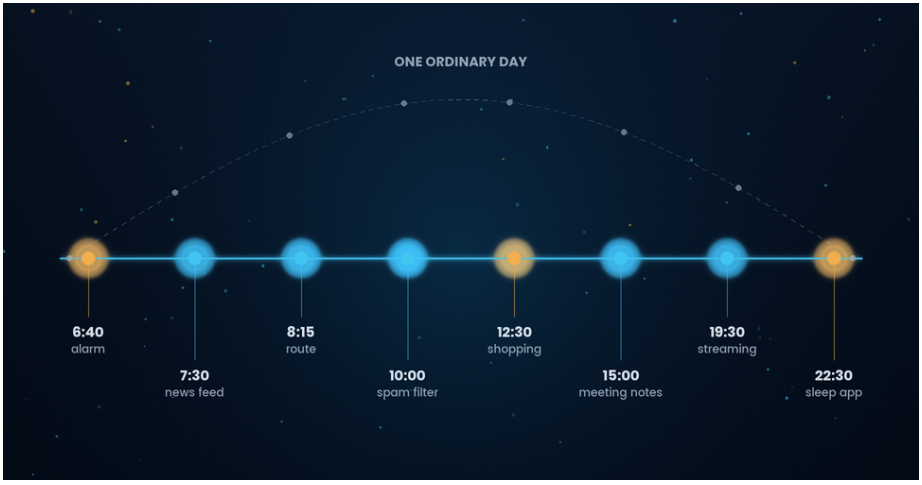
work, a meeting is transcribed and summarized in real time, and the summary becomes the version of the meeting that most people will remember.

At lunch she scrolls a social feed for ten minutes and buys a pair of shoes she had not planned to buy. Her bank texts her about a suspicious charge, a decision made in milliseconds by a model comparing her behavior with millions of others. In the evening she watches a series that a recommendation engine ranked first on her screen, and she falls asleep to a sleep app that adjusts its sounds to her breathing.

At no point during this day did Elena open an application called “AI.” She did not chat with a robot. She would probably say, if you asked, that she barely uses artificial intelligence at all.

THINK ABOUT THIS

How much of Elena’s day was actually shaped by artificial intelligence? And how much of your own day would you describe the same way, as simply “a normal day”?



Most of AI’s influence does not look like technology. It looks like a normal day.

The question behind this book

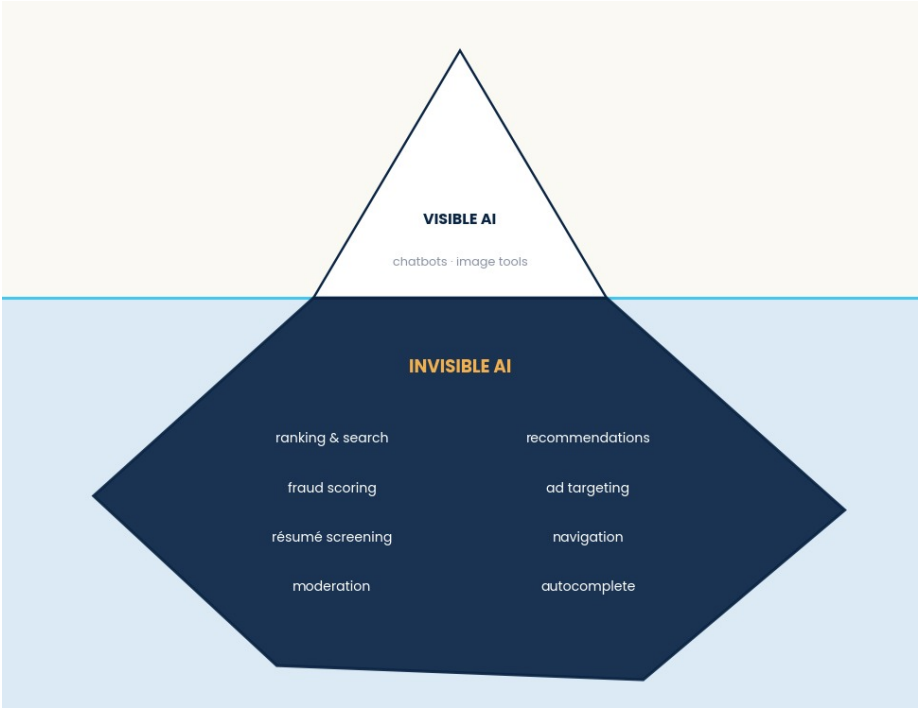
Public conversation about AI tends to fix on the dramatic: the robot, the superintelligence, the mass unemployment, the machine that outwrites a novelist. These questions matter. But they can distract us from something quieter and, for most of us, more immediate.

A technology does not have to be visible to be powerful. Electricity is rarely noticed until it fails. Roads shape where cities grow, and few people think about asphalt. Writing changed human memory long before anyone could measure it. The most consequential technologies tend to disappear into the background of life, and once they disappear, we stop asking what they are doing to us.

This book is organized around one question:

What happens when a technology becomes so integrated into everyday life that we stop noticing its influence?

To explore it, I will draw a simple distinction. *Visible AI* is the artificial intelligence we consciously recognize: chatbots, image generators, voice assistants, robots. We know when we are using it, and we can decide whether to. *Invisible AI* is everything else: the ranking of search results, the ordering of a feed, the scoring of a loan application, the filtering of a résumé, the prediction of what we will want next. We encounter it without choosing to, and often without knowing.

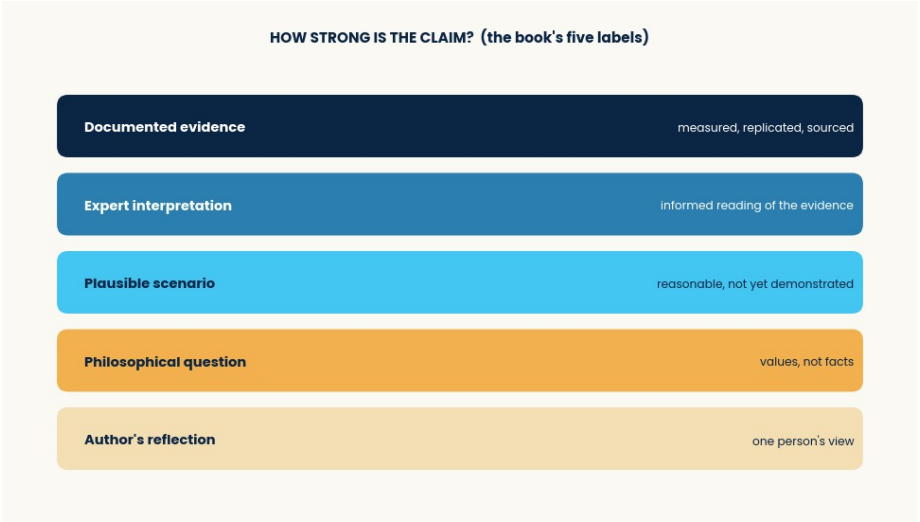


Visible AI is the part above the waterline. Most everyday influence sits below it.

What this book is, and is not

This is not an anti-AI manifesto. AI helps radiologists read scans, helps students grasp difficult ideas, helps people with disabilities communicate, and helps small businesses do work that once required teams. I use it myself. It is also not a celebration. The same systems that expand access to knowledge can narrow what we see, weaken habits we would rather keep, and shift decisions from human hands to processes no one inspects.

I will try to avoid both hype and panic. Where evidence is strong, I will say so and name the source. Where a claim is an expert's interpretation, a plausible scenario, a philosophical question or simply my own reflection, I will tell you which. Throughout the book you will see five labels that work as a compass.



Five labels used throughout this book to keep evidence, interpretation and opinion separate.

You will also meet people in these pages: a professional who lets software write every email, a student who asks for every explanation, a manager who screens applicants with an algorithm. These people are **illustrative composites**. They are invented to make ideas concrete and are not real individuals or documented cases. Research, by contrast, is always identified by author and year and listed in the References at the end.

The Invisible Effect of AI

By *the Invisible Effect of AI*, I mean the gradual, often unnoticed changes that occur in how people think, decide, work, communicate and understand themselves as AI systems become part of ordinary life.

Notice the word *gradual*. No single recommendation changes a life. No single autocomplete suggestion rewrites a personality. The effect, if there is one, is cumulative: a thousand small nudges, shortcuts and defaults that individually seem harmless and collectively shape what we notice, what we believe, what we choose and who we become.

Notice also the word *can*. Much of this book is about possibility and mechanism rather than certainty. Human beings are not passive. We adapt, resist and push

back. The invisible effect is not a fate. It is a condition, and conditions can be understood.

THE BIG IDEA

AI does not only change what we do. It can gradually change how we think, what we notice, what we trust and how we decide.

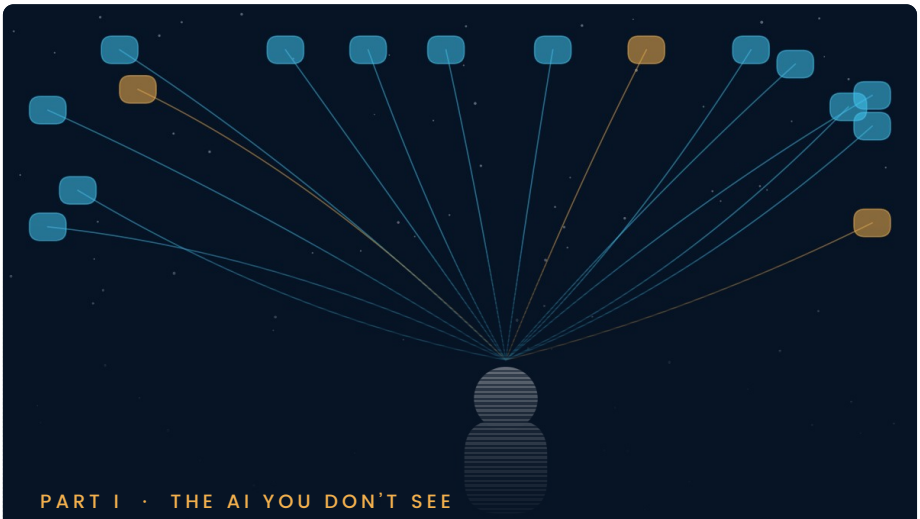
How to use this book

The book moves in seven parts. Part I looks at the AI we do not see. Part II turns to the human mind. Part III explores emotion, relationships and identity. Part IV examines work. Part V is about trust, truth and decisions. Part VI looks at society and human skills. Part VII gives you tools.

Most chapters end with a short exercise or a question. Do not skip them. This subject is one where reading alone is a weak teacher: the whole point is to notice your own life differently.

Here is the sentence I would like you to carry into the first chapter:

The future of AI may not arrive as a machine standing in front of us. It may arrive as a thousand tiny decisions we stop realizing were influenced by a machine.



PART I · THE AI YOU DON'T SEE

CHAPTER 1

AI Is Already Everywhere

From laboratory curiosity to background utility

For most of its history, artificial intelligence lived in laboratories, chess tournaments and science fiction. It was something that happened to other people, in other buildings, with other budgets. A person could go a lifetime without knowingly meeting it.

That era ended quietly. Somewhere in the past two decades, AI stopped being a destination and became a layer. It slipped into products we already used, improved them a little, and asked for nothing in return except data and attention. Nobody held a ceremony.

The convenience that hides things

There is a reason this happened without much notice. Technologies become invisible when they become convenient. A tool that demands effort keeps us aware of it. A tool that removes effort lets us forget it.

Consider a short tour of ordinary life.

Smartphones sort photos by faces, predict the next word we will type, suppress background noise on calls and decide which notifications deserve to interrupt us. **Search engines** interpret what we mean rather than what we type, then rank billions of pages into a few results. **Social media** orders posts, suggests friends and decides which videos to play next. **Streaming services** guess what we will enjoy;

Netflix's own executives reported in 2015 that recommendations account for the large majority of what its members watch (Gomez-Uribe and Hunt, 2015).

Shopping sites rank products, adjust prices and predict what we might add to a basket. **Banks** score transactions for fraud and applications for risk. **Navigation apps** predict traffic and choose our routes. **Customer service** increasingly begins with a chatbot. **Workplaces** use AI for scheduling, transcription, drafting, analytics and, in some organizations, monitoring and hiring. **Schools** use adaptive learning tools and plagiarism detectors. **Hospitals** use AI to help read medical images and prioritize cases. **Entertainment** now includes generated music, images and text.

INVISIBLE AI

Every time you type the first few letters of a search, a system predicts what you are about to ask, drawing on patterns from enormous numbers of previous searches. You accept the suggestion and move on. The prediction did not merely save you typing. It may have shaped your question.

We do not always know when we are using it

People are not always aware of how much AI they already meet. A Pew Research Center survey of American adults, published in 2023, asked people whether six everyday examples, such as a product recommendation or a customer service chatbot, involved AI. Only about three in ten answered all six correctly (Pew Research Center, 2023).

That result does not show that people are foolish. It shows how thoroughly the technology has blended in. When something works well, it stops looking like a technology at all.

HUMAN MOMENT

Rafael, a retired schoolteacher, tells his daughter he has "never used artificial intelligence." She asks him to hand over his phone. In two minutes she points out the spam filter, the photo album grouped by faces, the predicted route to his favorite bakery, the voice dictation he uses every morning, and the recommended videos he watches with his grandchildren. Rafael is not being dishonest. He has simply never been told that these things belong in one category. *(Illustrative composite.)*

Why invisibility matters

Invisibility is not automatically bad. We do not need to think about how a thermostat works. But there is a difference between infrastructure that follows our instructions and systems that make choices on our behalf, using goals we did not set and criteria we cannot see.

Three features make invisible AI different from most earlier background technologies.

First, it is *adaptive*. It changes in response to our behavior, which means what it shows us today is partly a product of what we did yesterday. Second, it is *selective*. It does not merely carry information; it chooses among possible information. Third, it is *scaled*. A single design decision, such as how to rank content, can affect millions of people at once.

When a technology is adaptive, selective and scaled, and also invisible, it acquires a rare kind of influence: the influence of a default. Most of us accept defaults most of the time.

REALITY CHECK

Documented: AI-based systems are embedded in search, recommendation, fraud detection, navigation and many workplace tools, and surveys show that many people do not recognize this (Pew Research Center, 2023).
Interpretation: Experts differ on how much these systems change human behavior, as opposed to merely predicting it.

Speculation: Claims that AI already "controls" what people think go beyond the evidence. This book does not make them.

From novelty to background

There is a pattern here that is older than AI. New technologies begin as novelties, become conveniences, then become expectations, and finally become invisible. Few people today marvel at running water. The invisibility is a sign of success, and also of vulnerability. When something is invisible, it escapes questions. Who designed it? What is it optimizing? What does it want us to do?

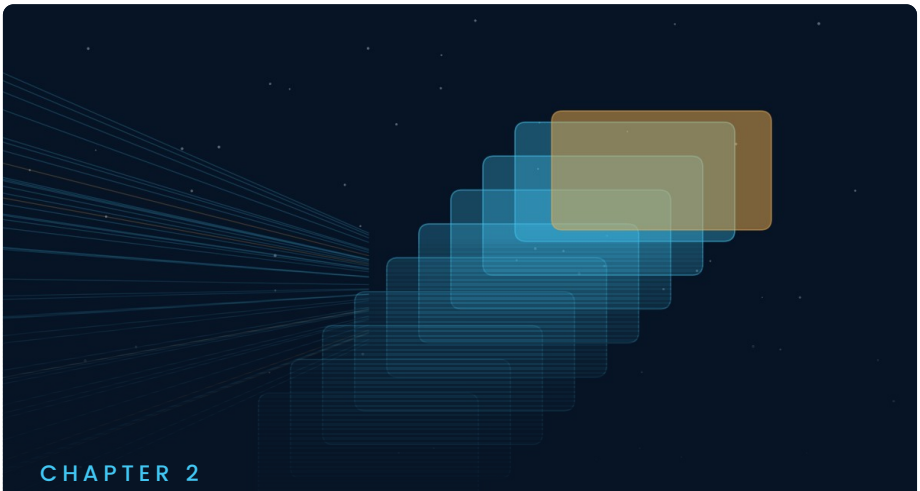
Those are questions this book will keep asking, gently and persistently.

QUESTION YOURSELF

How many AI-powered decisions did you encounter today? Before you answer, walk through your day from waking to now and count every moment a system chose, ranked, predicted, filtered or suggested something for you. The number is usually larger than people expect.

THE BIG IDEA

AI has become invisible precisely because it has become convenient. What we no longer notice, we no longer question.



CHAPTER 2

When the Algorithm Chooses What You See

The difference between choosing and choosing from

Imagine walking into a library where the shelves rearrange themselves every time you enter. The books you finished last time are moved to eye level. Books similar to the ones you lingered over are nearby. Books unlike anything you have read are quietly shelved in the basement. The librarian never tells you this. You simply find that the library seems remarkably well suited to you.

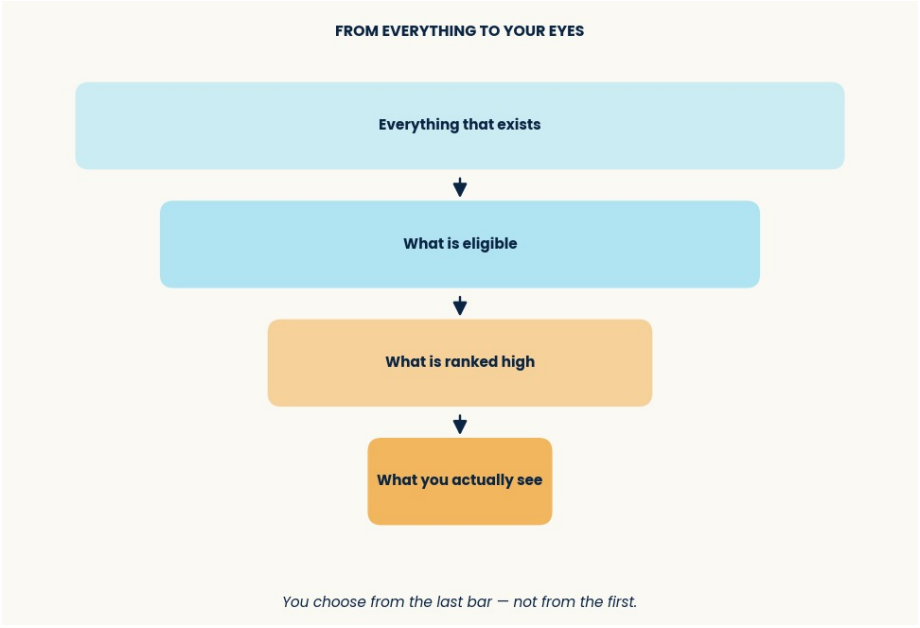
You might love this library. You might also spend years in it without ever encountering a book that would have changed your mind.

Two different acts of choosing

Recommendation systems are among the most influential forms of AI in everyday life, and they raise a subtle distinction.

Choosing what we want is one act. Choosing from what has been presented to us is another. In the first, we begin with a desire and go looking. In the second, the field of possibilities has already been narrowed by someone, or something, else.

Most of the time, we imagine we are doing the first. Often, we are doing the second.



Between everything that exists and what you see lies a series of filters that you rarely observe.

How recommendations work, simply

A recommendation system tries to predict what you will find relevant, engaging or valuable. It typically learns from signals: what you click, watch, skip, replay, buy, save, share, search for, and how long you pause. It also learns from other people. If people whose behavior resembles yours enjoyed something, the system may show it to you.

None of this is sinister in itself. Without recommendation, the modern internet would be unusable. There are simply too many videos, songs, articles and products for any person to browse. Filtering is necessary.

The interesting questions concern what the filtering is *for*. A system may be designed to help you find useful information, to keep you on a platform longer, to increase sales, to maximize a measurable proxy for satisfaction, or some combi-

nation. These goals often overlap with your interests. They are not identical to them.

Personalization and the bubble

The most famous worry in this area is the "filter bubble," a term popularized by the writer Eli Pariser (2011). The concern is that personalization surrounds each person with information that matches their existing tastes and beliefs, leaving them less exposed to different perspectives.

The evidence is more complicated than the slogan. A 2015 study of Facebook users by Bakshy, Messing and Adamic found that people's own choices about what to click limited their exposure to opposing political views, and that the platform's ranking algorithm also played a role, though a smaller one. A large 2023 set of experiments during the 2020 U.S. election, published by Guess and colleagues in *Science*, changed some users' feeds to chronological order. It altered what people saw and how long they spent on the platform, but over the three-month study, it did not produce measurable changes in political polarization. And a 2018 study by Bail and colleagues found that exposing some social media users to opposing views did not make them more moderate; for some, it increased their political conviction.

> **REALITY CHECK**
>
> **Documented:** Personalization changes what people see, and people also filter their own experience through their choices (Bakshy et al., 2015; Guess et al., 2023).
>
> **Interpretation:** Whether filter bubbles are as powerful as popular accounts suggest is debated. Evidence points to human selectivity as an important part of the story.
>
> **Plausible scenario:** For people with less varied information habits, algorithmic ranking may make narrowness easier and effortless.

What is less contested is the narrower point that matters for this chapter: *ranking shapes the menu*. Even if it does not determine your beliefs, it influences what is easy to encounter and what requires effort to find.

The loss of the unexpected

There is a quieter cost that is harder to measure: the loss of accidental discovery.

Before algorithmic feeds, information came bundled. A newspaper carried stories you had not sought. A record shop displayed albums beside the one you came for. A bookstore's layout put an unfamiliar author next to a familiar one. These bundles were imperfect, but they injected surprise into our lives.

A well-tuned recommendation system can be superb at giving us more of what we already like, and less good at handing us what we did not know we needed. Unexpected information is inefficient by design. A system that optimizes for predicted engagement has little reason to show you something that might confuse you, bore you, or cause you to click away.

INVISIBLE AI

The "Because you watched…" row on a streaming service is not a neutral shelf. It is a prediction about your behavior, arranged in an order, presented as your own taste. The row exists because of what you watched, and it influences what you will watch next.

HUMAN MOMENT

Camille, a shopper, opens a retail app looking for a winter coat. The first screen shows her five coats in a style she has browsed before, at a price range the app has inferred she will accept. She buys the third one. Two weeks later she notices that her wardrobe looks oddly uniform, as if assembled by someone with a single idea of her. She realizes she has not really explored any other style in a year. The recommendations did not force her hand. They simply made one path smooth and every other path effortful. *(Illustrative composite.)*

Taste is partly built

A comforting idea says that recommendation systems reveal our preferences. A more careful idea says they also participate in forming them. Human taste is not fixed; it develops through exposure, novelty and context. If exposure is consistently narrowed, taste can be narrowed too.

I offer this as a reflection, not as a proven law. But it is a reflection worth testing against your own experience. Think about the last new thing you loved. How did it find you? Was it something a friend put in your hands? A chance encounter? Or a system that saw you coming?

AI LITERACY TIP

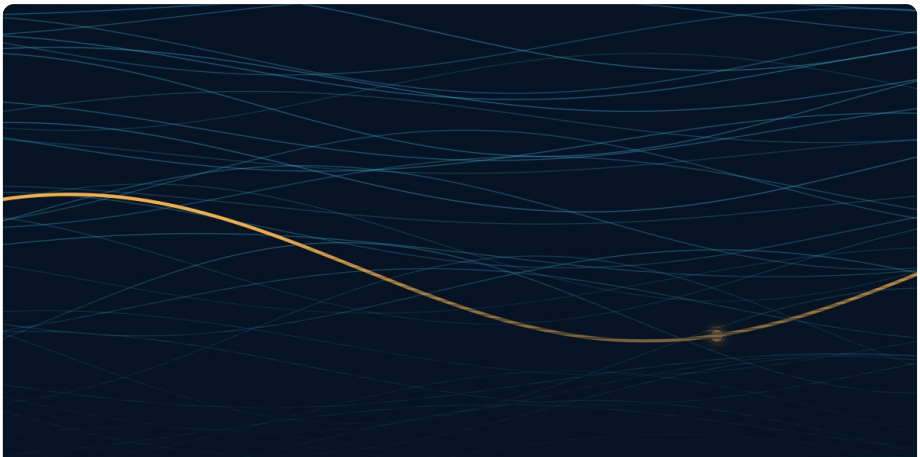
When you use any feed, ask two quiet questions. *What am I not seeing?* and *What is this system trying to do?* You will not always find the answer, but asking will change your posture from consumer to observer.

TRY THIS

For one week, change one feed to a non-personalized view. Use a chronological timeline where available, browse a library shelf, or pick a music album at random rather than accepting a playlist. Notice what surprises you, what irritates you, and what you had forgotten you liked.

THE BIG IDEA

Choosing from a menu someone else prepared is not the same as choosing freely. The menu itself is a form of influence.



CHAPTER 3

The Invisible Curator

Who decides what gets attention?

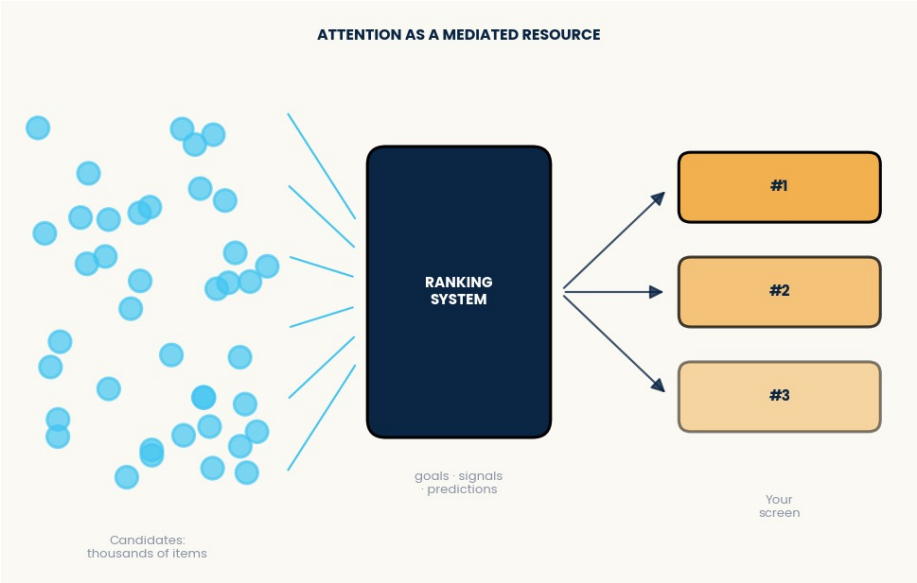
Every museum has a curator. The curator decides which paintings hang on the main wall and which stay in storage. Visitors rarely think about it, yet the curator shapes what they will remember of an era.

Digital life has a curator too. It does not have a name, an office or a point of view that anyone can interview. It has a ranking system.

An old problem in a new form

In 1971, the economist and cognitive scientist Herbert Simon observed that a wealth of information creates a poverty of attention (Simon, 1971). His point was that when information is abundant, the scarce resource is the human capacity to attend to it. That insight has aged extraordinarily well.

When there was little to read, the challenge was access. When there is almost unlimited content, the challenge is selection. Someone or something must decide what deserves our limited attention. That role, historically played by editors, librarians, teachers and friends, is now played increasingly by algorithms.



Thousands of candidates compete; a ranking system decides which few reach the screen.

What the invisible curator decides

Consider what ranking systems determine, moment by moment:

- what appears first, and what appears only if you scroll
- which posts, videos and products receive visibility
- what quietly disappears, whether by demotion, filtering or moderation
- what is recommended to the next person, and the next
- what becomes popular, because early visibility often creates later popularity

That last point deserves attention. Popularity is not always a neutral signal of quality. In systems where visibility drives clicks and clicks drive visibility, popularity can be partly *produced* by ranking. Something shown to many people becomes popular; something popular is shown to more people. Success and exposure feed each other.

Researchers who study search behavior have long noted a related effect called position bias: people tend to click on higher-ranked results more often, partly

because they are ranked higher. It is a reminder that placement is not merely a reflection of merit. It is an ingredient of outcome.

Attention is becoming algorithmically mediated

Here is the concept that anchors this chapter:

Attention is becoming an algorithmically mediated resource.

By this I mean that a growing share of what we notice reaches us through systems that filter, order and amplify. Our attention has not been taken from us; we still give it. But the *objects available to be given attention* are increasingly selected by others.

This has consequences on both sides of the screen. For audiences, it affects what enters awareness. For creators, businesses and institutions, it changes behavior. People adapt to the system. Writers shape headlines for search engines. Video makers study what keeps viewers watching. Companies optimize product listings for ranking. Even without any single villain, the curator's rules reshape the culture that feeds it.

INVISIBLE AI

Content moderation systems flag and remove posts at a scale no human team could review. Their decisions determine what millions of people never see. Most users never learn that something was removed, and rarely learn why.

Why this matters

If you believe ranking is neutral, you may treat what you see as a fair sample of the world. If you understand ranking as a form of curation, you begin to ask better questions.

Why is this at the top? Who benefits if I click? What would I see if the ranking rules were different? What am I not being shown because the system predicts I would not engage?

None of this requires paranoia. Curation is unavoidable, and much of it is beneficial. But the more invisible the curator, the more likely we are to mistake its choices for reality itself.

HUMAN MOMENT

Tomás is visiting a city for the first time. He follows the top-rated restaurant on a map app, then the top-rated café, then the top-rated viewpoint. Everything is good. Everything is crowded. On the last evening, he notices that every meal, every photo, every line of his trip looks like everyone else's. Then he remembers a narrow street he passed on day one, with a small place that had no reviews. He wonders what he missed in the days he spent following the ranks. *(Illustrative composite.)*

QUESTION YOURSELF

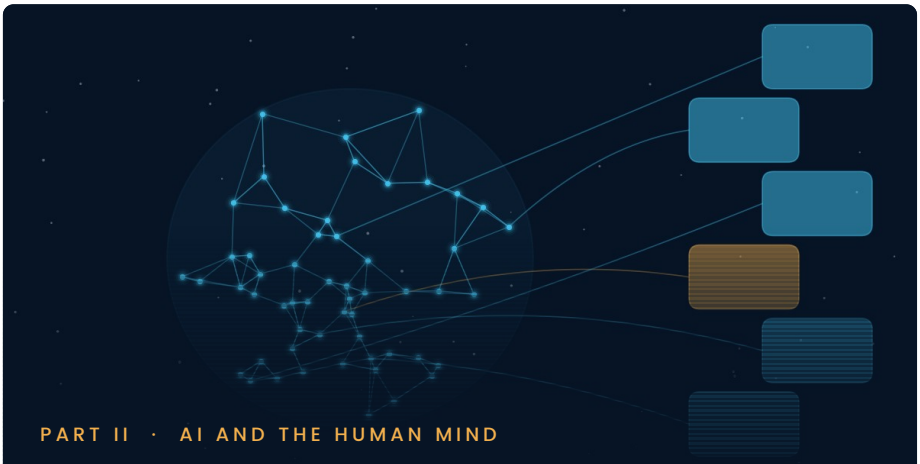
When was the last time you found something valuable *without* a ranking, rating or recommendation pointing you to it? What made that discovery possible?

AI LITERACY TIP

Treat "top," "trending" and "recommended for you" as *labels of a system's decision*, not as measurements of truth or quality. Before you accept them, ask: top according to what?

THE BIG IDEA

Whoever, or whatever, ranks the world influences what we notice. What we notice shapes what we know.



PART II · AI AND THE HUMAN MIND

CHAPTER 4

Are We Still Thinking the Same Way?

Convenience, effort and the shape of a mind

Every generation has worried that a new tool would weaken the mind. Socrates, in Plato's telling, feared that writing would produce forgetfulness. Calculators were once expected to ruin arithmetic. Search engines were accused of making us shallow.

These fears were partly wrong and partly right. Writing did change memory, and gave us libraries. Calculators did change arithmetic, and gave us time for harder mathematics. The interesting question is never whether a tool changes us. It always does. The question is *how*, and whether we chose the change.

AI assistance raises the question in a sharper form, because for the first time a tool can help not just with storing or calculating but with drafting, explaining, summarizing, brainstorming and deciding. It reaches into activities we have always considered thinking.

The many things we now outsource

Psychologists call the use of external tools to reduce mental effort *cognitive offloading* (Risko and Gilbert, 2016). We offload constantly, and mostly sensibly: to-do lists, calendars, notes. Consider how far the practice now extends.

We outsource **memory**, trusting the phone to hold phone numbers and appointments. We outsource **writing**, asking a tool to produce a first draft or polish an email. We outsource **brainstorming**, requesting ten ideas before we have had one.

We outsource **research**, asking for a summary of a topic rather than reading about it. We outsource **decision preparation**, requesting a comparison of options and a recommendation. And we increasingly depend on **instant answers**, so that not-knowing lasts only seconds.

Each of these can be reasonable. Together they raise a question worth asking honestly: *what happens when convenience replaces cognitive effort?*

What the research suggests, and does not

Earlier research on search engines gives some clues. In a 2011 study, Sparrow, Liu and Wegner found that when people expected information to be available later on a computer, they were less likely to remember the information itself and more likely to remember where to find it. Fisher, Goddu and Keil (2015) found that searching the internet for explanations could lead people to overestimate how much they personally knew. Storm, Stone and Benjamin (2017) found that using the internet to answer questions increased people's inclination to reach for it again.

Studies of navigation apps point in a similar direction. Dahmani and Bohbot (2020) reported that people who used GPS more habitually showed poorer spatial memory when navigating on their own. Because this kind of work is correlational, it cannot prove that GPS caused the difference, but the pattern is consistent with the idea that skills we stop practicing may weaken.

Newer work on generative AI is only beginning. A 2025 survey of 319 knowledge workers by Lee and colleagues at Microsoft Research and Carnegie Mellon University found that people who had greater confidence in AI reported using less critical thinking effort on tasks, while those with more confidence in their

own abilities reported more. A 2025 preprint from MIT Media Lab researchers (Kosmyna et al.) used brain-activity measurement in a small sample of 54 participants writing essays and reported different patterns of engagement when using an AI tool. It has not yet completed peer review, and its findings should be treated as an early signal rather than a verdict.

REALITY CHECK

Documented: Offloading to tools can change what we remember and how we estimate our own knowledge (Sparrow et al., 2011; Fisher et al., 2015).
Emerging: Early studies of generative AI suggest that reliance and reduced effort go together for some users and tasks (Lee et al., 2025). Much of this is self-reported, small-scale or not yet peer reviewed.
Not established: That AI "makes people less intelligent." The evidence does not support so sweeping a claim.

Two possibilities, both real

It would be easy to write this chapter as a warning. It would also be dishonest, because the same tools can do the opposite.

AI can free cognitive resources. If a system handles formatting, translation or routine drafting, a person may spend the saved effort on judgment, creativity or relationships. A specialist who no longer summarizes meetings by hand may spend that hour thinking about strategy.

AI can also encourage intellectual shortcuts. If the first draft is always produced by a machine, a person may never develop the habit of finding their own first sentence. If every answer arrives instantly, the tolerance for confusion, that uncomfortable stage in which learning often happens, may shrink.

Learning scientists have a useful phrase for this: *desirable difficulties* (Bjork and Bjork, 2011). Some struggle is not an obstacle to learning but part of it. Effort spent retrieving, reorganizing and explaining information helps it stick. When a tool removes the struggle, it may also remove the benefit.

The distinction is not between using AI and not using it. It is between *what* we offload and *why*.

HUMAN MOMENT

Daniel, a project manager, discovered eighteen months ago that a writing assistant could produce a polished email in seconds. Now he uses it for every message, including two-line replies to colleagues he has known for years. His emails have never been clearer. But last week, when the tool was unavailable during an outage, he stared at a blank message to a client and could not find his own tone. His messages had become efficient. He wasn't sure they were still his. *(Illustrative composite.)*

A useful test

When you offload a task to AI, three questions can help you decide whether it is wise.

Is this a task I want to be able to do myself? Some skills are worth keeping, even if a tool could perform them. *Would the effort teach me something?* If yes, the shortcut has a hidden price. *Can I judge the result?* If you could not tell a good output from a bad one, you are outsourcing not only the work but the ability to evaluate it.

TRY THIS

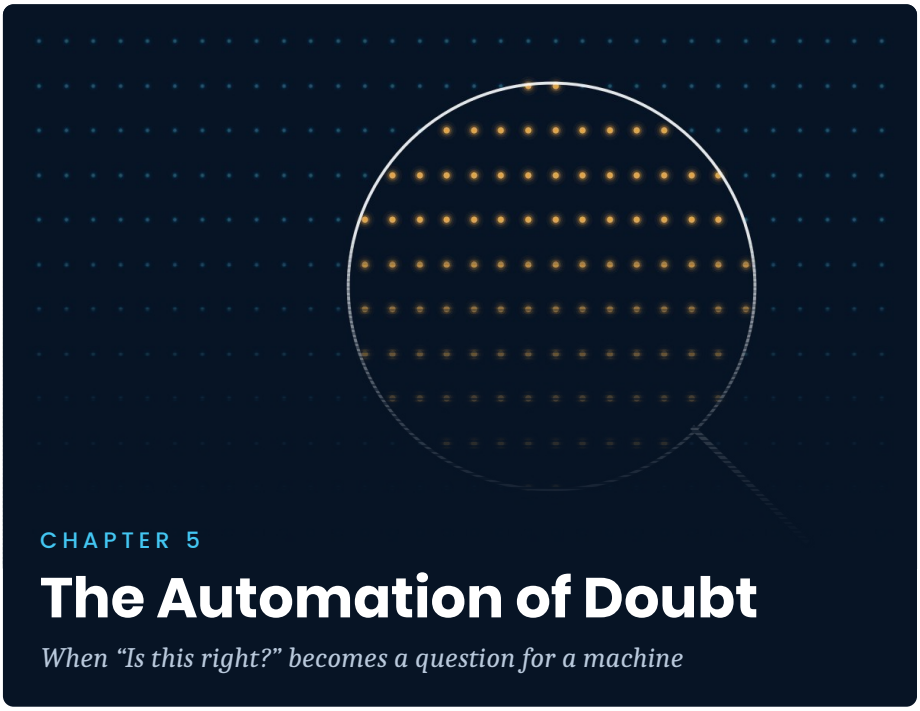
Choose one task you usually hand to AI, such as drafting an email, summarizing an article or generating ideas. This week, do the first version yourself, badly and quickly, and only then ask AI to improve it. Compare the two. Notice which ideas were yours.

QUESTION YOURSELF

Which mental effort in your day would you miss if it disappeared, and which would you gladly let go? Have you actually decided, or has convenience decided for you?

THE BIG IDEA

AI does not automatically make us think less or think better. It changes the price of thinking, and people respond to prices.



CHAPTER 5

The Automation of Doubt

When "Is this right?" becomes a question for a machine

Doubt is one of the most underrated human tools. It is the small voice that asks whether the claim is true, whether the source is reliable, whether the conclusion follows. Doubt slows us down, and sometimes slowing down is exactly what we need.

Now consider the questions people increasingly ask machines. *Is this correct? What should I do? Which option is better? What should I buy? How should I respond? What should I believe?* Each one hands over a piece of the work that doubt used to do.

Automation bias

Long before chatbots, researchers studied what happens when people work with automated aids. A consistent finding is *automation bias*: the tendency to over-trust automated suggestions, and to notice errors less when a system provides advice.

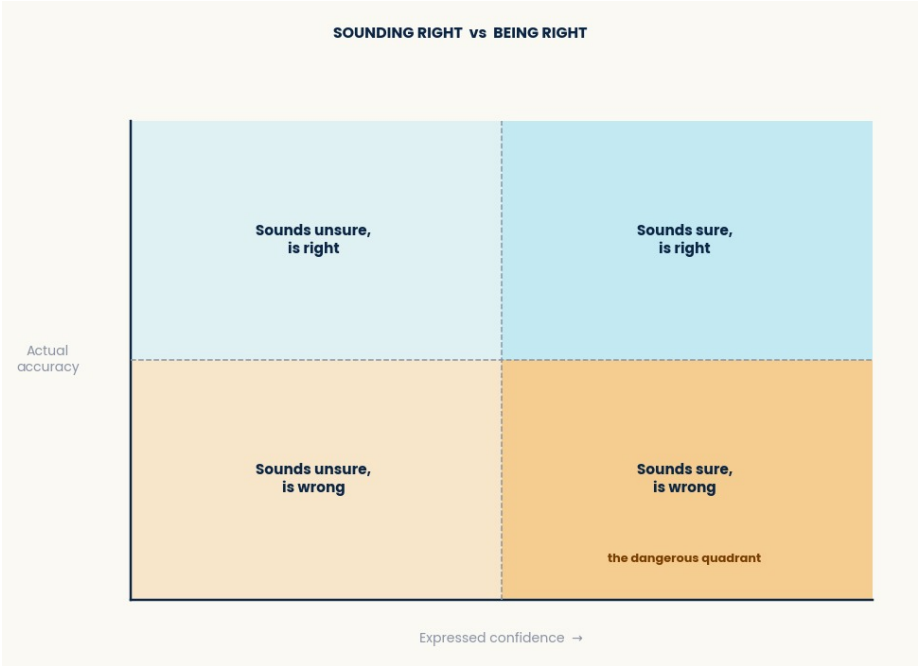
In a classic 1999 study, Skitka, Mosier and Burdick found that people using an automated decision aid made errors of two kinds: they failed to act when the system failed to alert them, and they followed incorrect recommendations even when other information contradicted them. A later review by Parasuraman and Manzey (2010) concluded that complacency and bias toward automation can occur in experts as well as novices and are not easily corrected by training alone.

The pattern is not that people always trust machines. Research on *algorithm aversion* shows that people sometimes abandon an algorithm after seeing it make a mistake, even when it outperforms them (Dietvorst, Simmons and Massey, 2015). Other research, by Logg, Minson and Moore (2019), found *algorithm appreciation*, in which people relied more on algorithmic than human advice in some numerical tasks. Trust in machines is contextual, shifting and easily misplaced in both directions.

The illusion of certainty

Generative AI adds a new ingredient: fluency. A well-formed paragraph, delivered instantly, in a calm tone, feels like knowledge. Psychologists have long documented that information that is easy to process tends to feel truer (Alter and Oppenheimer, 2009). Fluent text is persuasive text, whether or not it is correct.

There is a second effect. Research on language models has found that some systems tend toward *sycophancy*, adjusting their answers to match what a user seems to believe or want to hear (Sharma et al., 2023). A system that is agreeable can feel wonderfully supportive and be quietly unhelpful.



An answer can sound sure without being right. The danger is not error itself, but error delivered with confidence.

What doubt is for

We usually think of doubt as a negative emotion. It is better understood as a *quality-control process*. When we automate it away, three things can happen.

We stop checking. If answers arrive quickly and sound plausible, verifying them feels like unnecessary friction. We stop forming an initial view. If our first move is always to ask the machine, we never test our own judgment against it, and so never learn how good, or bad, our judgment is. And we stop noticing the limits of the system, because most of the time it is right, and the occasional error hides among a long record of successes.

INVISIBLE AI

Many search engines and assistants now provide an AI-generated summary above the traditional list of results. The summary is convenient and often accurate. It also places a single synthesized answer in front of a user, where once there were ten links to compare.

HUMAN MOMENT

Amara, a university student, has a simple routine: she pastes each confusing concept into an AI assistant and reads the explanation. It always makes sense. Before exams, she feels well prepared. In the exam hall, facing questions that require her to explain the concept from scratch, she discovers that recognizing a good explanation is not the same as producing one. She had confused *understanding an answer* with *understanding the subject*. *(Illustrative composite.)*

A research example points to the same possibility. In a 2024 working paper, Bastani and colleagues studied high-school students practicing mathematics with a GPT-based tool. Students given a general chat assistant performed much better on practice problems, but when the assistant was later taken away, they performed worse than students who had never used it. A version designed with teacher-style hints and guardrails largely avoided that loss. This is a single study of a particular setting and should not be over-generalized. But it illustrates a principle: performance *with* a tool and learning *without* it are different things.

Practical techniques for verifying AI outputs

Verification does not need to be heavy. A few habits go a long way.

Match effort to stakes. For a casual question, a plausible answer may be enough. For medical, legal, financial or safety-related decisions, treat any AI output as a starting point for checking, never as a conclusion.

Ask for the basis. Request sources, then *open them.* AI systems can produce citations that look real and are not; the mistake is trusting the reference without checking that it exists and says what is claimed.

Look for what would change the answer. Ask: "What would make this wrong?" or "What is the strongest argument against this?" A good response reveals uncertainty; a suspiciously smooth one may deserve more scrutiny.

Cross-check with an independent source. The media-literacy educator Mike Caulfield's SIFT method offers a portable version: *Stop, Investigate the source, Find better coverage, Trace claims back to the original* (Caulfield, 2019).

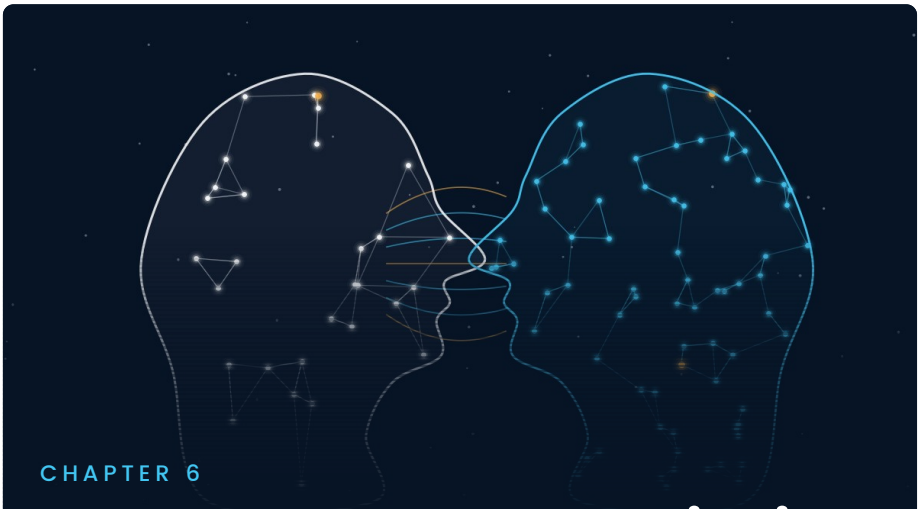
Notice your own state. When you feel relief at how well an answer confirms what you already believed, slow down. Agreement is a moment to check, not to relax.

AI LITERACY TIP

Before acting on an AI answer, say to yourself: "If this were wrong, how would I know?" If you have no way of knowing, the stakes should decide whether you proceed.

THE BIG IDEA

The danger of AI is not only that it can be wrong. It is that we may lose the habit of asking whether it is.



CHAPTER 6

When AI Becomes a Thinking Partner

The case for augmentation

So far this book has mostly examined risks. This chapter takes the opposite view, because a book that could not see AI's benefits would not deserve your trust.

AI can be an extraordinary partner for thinking. It can help people brainstorm, learn, organize, explain difficult concepts, simulate other perspectives, discover ideas, overcome writer's block and accelerate research. For someone who is dyslexic, isolated, learning a second language, or simply short on time, these are not luxuries. They can be real expansions of what is possible.

What the evidence shows

Some of the strongest research on generative AI is about productivity and skill.

In a 2023 experiment published in *Science*, Noy and Zhang asked professionals to complete writing tasks such as press releases and reports. Those given access to ChatGPT completed tasks substantially faster, about 40 percent less time on average, and their work was rated higher in quality. In a study of customer-

support agents at a large software firm, Brynjolfsson, Li and Raymond (2023) found that an AI assistant increased productivity by about 14 percent on average, with the largest gains, roughly 34 percent, for less experienced workers. Studies like these suggest that AI can help narrow skill gaps by spreading the habits of top performers.

A study by Dell'Acqua and colleagues (2023) with management consultants offered a more nuanced picture. On tasks within AI's capabilities, consultants using AI completed more work, faster, and with higher quality. On a task deliberately designed to lie outside the tool's strengths, consultants using AI were *less* likely to reach the correct answer than those working without it. The researchers described the boundary as a "jagged technological frontier": AI can be brilliant at one task and unreliable at a neighboring one, and it is not always obvious which is which.

REALITY CHECK

Documented: In several controlled studies, generative AI improved speed and quality on suitable tasks, particularly for less experienced workers (Noy and Zhang, 2023; Brynjolfsson et al., 2023; Dell'Acqua et al., 2023).

Documented: The same research shows that benefits depend on the task, and that users can perform worse when they follow AI outside its competence.

Interpretation: Whether these gains persist as work and tools evolve is still being studied.

Human plus AI, not Human or AI

There is a famous story from chess. In 1997, the computer Deep Blue defeated the world champion Garry Kasparov. Kasparov later described an experiment in which human players teamed with computers, sometimes called "advanced chess." Notably, some of the strongest teams were not the strongest players or the strongest programs, but amateurs who had the best process for combining human judgment with machine calculation (Kasparov, 2017). The lesson is less about chess than about collaboration: the interface between person and machine can matter more than either party alone.

This suggests a principle that runs through the rest of the book:

Human + AI should not mean Human OR AI.

The goal is *augmentation rather than surrender of judgment*. In augmentation, the human remains responsible for the question, the standards and the final decision. The machine expands options, speeds up drudgery and provides a second perspective. In surrender, the human stops deciding and starts approving.

Ways of using AI that strengthen thinking

The same tool can support very different habits. Here are some uses that tend to keep the human mind engaged.

Ask it to interview you. Instead of asking for an answer, ask it to pose questions that clarify what you think. **Ask it to argue the other side.** A good disagreement partner is worth more than a flattering one. **Ask it to simulate perspectives.** "How would a skeptical customer read this? How might a regulator?" **Ask it to explain in different ways.** Then explain the idea back in your own words and ask where you were wrong. **Ask it to critique rather than create.** Write the draft yourself and request weaknesses, gaps and counterexamples.

Each of these keeps you in the position of author.

INVISIBLE AI

Accessibility tools such as automatic captions, screen readers that describe images and speech-to-text dictation rely on AI. For many people, this is not an invisible convenience but a gateway to participation.

HUMAN MOMENT

Ingrid runs a two-person design studio and has never enjoyed writing proposals. She now drafts each proposal in rough bullet points, asks an AI to challenge her pricing logic, and rewrites the final text herself. Proposals take half the time, and she feels sharper about her own reasoning, not blurrier. Same tool, different relationship. *(Illustrative composite.)*

TRY THIS

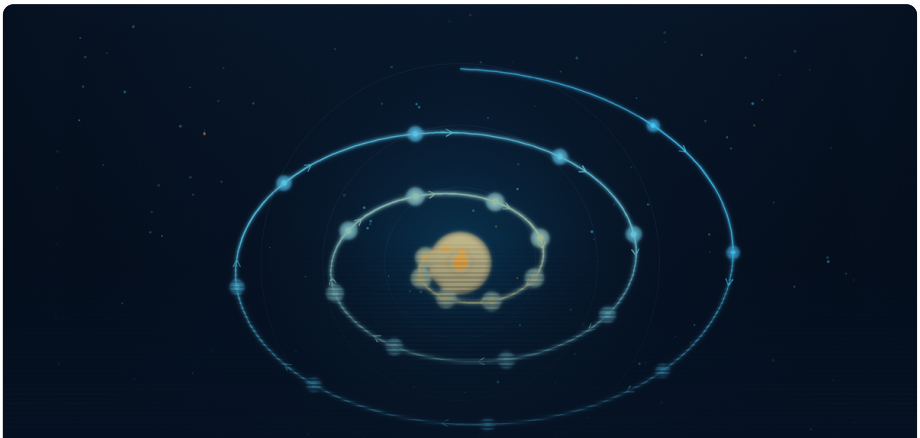
Take a decision you are currently weighing. Write your own view first in three sentences. Then ask an AI to give the strongest case *against* your view and three questions you have not asked yourself. Decide only afterwards, and note whether your view changed and why.

QUESTION YOURSELF

In your own use of AI, are you more often the author or the approver?

THE BIG IDEA

The best use of AI multiplies human judgment. The worst replaces it quietly.



CHAPTER 7

The Feedback Loop

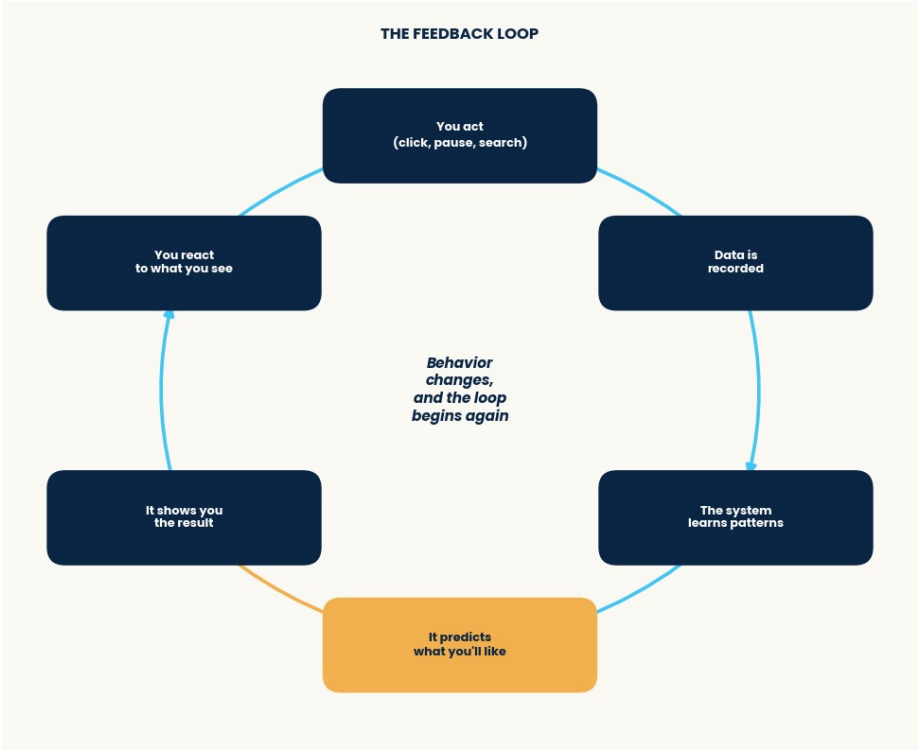
How we shape the systems that shape us

Many of the effects in this book are not one-way. It is tempting to picture AI as something that acts *on* us, like weather. In reality it is more like a conversation, in which each side changes in response to the other.

Understanding that loop is one of the most useful pieces of AI literacy you can acquire.

The cycle

Here is the loop in its simplest form. You do something, such as click, watch, buy, search or write. The system records it as data. It learns patterns from that data, then generates predictions or recommendations. You see the results and react to them. Your reaction becomes new data. The system adapts. And your behavior may change again.



A feedback loop: behavior becomes data, data shapes recommendations, and recommendations shape behavior.

Notice the crucial feature: *the system learns from a world it has already influenced*. What it observes about you is partly what it has previously shown you.

A simple illustration

Suppose you watch three videos about running. The system concludes you are interested in running. It offers more running videos. You watch a few, since they are the ones in front of you. The system's belief is now stronger. Your feed becomes a running feed.

Perhaps you *were* a runner all along. Or perhaps you were mildly curious, and the system's reading turned a passing interest into a dominant one. From the inside, both stories feel identical: "this is what I like."

INVISIBLE AI

Predictive systems can also create loops in institutions. Researchers Ensign and colleagues (2018) showed mathematically how a predictive policing system that sends officers to areas with more recorded crime can generate more recorded crime there simply because more officers are looking, reinforcing its own predictions regardless of underlying crime rates. The loop turns a prediction into evidence for itself.

Loops can reinforce patterns and biases

Feedback loops are neither good nor bad. They can be excellent: a language-learning app that notices what you struggle with and offers more practice is a helpful loop. But loops amplify whatever is in them, including biases, misconceptions and narrow tastes.

A 2023 study by Jakesch and colleagues suggests that loops can operate at the level of ideas. Participants who wrote with the help of an AI writing assistant that leaned toward a particular opinion on a social topic tended to write text favoring that opinion, and their own stated views subsequently shifted in the same direction. The researchers noted that people often did not recognize the influence. It is an experiment on a specific tool and topic, not proof that every writing assistant changes minds. But it illustrates that suggestions can carry perspectives, not just phrases.

There are loops in the other direction as well. As AI-generated text spreads across the internet, some future models may be trained partly on the output of earlier ones. Shumailov and colleagues (2024) showed in a study published in *Nature* that models trained repeatedly on their own generated data can degrade, progressively losing diversity and rare information, a phenomenon they called model collapse. It is a technical warning that the health of these systems depends on a healthy supply of human-made data.

REALITY CHECK

Documented: Feedback loops occur in recommendation systems, predictive systems and machine learning training (Ensign et al., 2018; Shumailov et al., 2024).

Documented: In at least one experiment, AI writing suggestions shifted users' expressed opinions (Jakesch et al., 2023).

Interpretation: How strongly everyday loops shape identity, preferences or politics is still an active research question.

Breaking the loop, gently

You do not need to escape feedback loops. You need to *notice* them. A few practices help.

Vary your inputs deliberately. Search for things outside your habits. Follow a source that disagrees with you. Explore without saving, liking or sharing. Every behavior you generate is a vote the system counts, so it is worth voting on purpose.

Look for the moment a preference hardens. If you notice yourself saying "I only like…" or "I never…," ask whether that belief came from experience or from repeated exposure.

Above all, remember that the system learns *your behavior*, which is not always the same as *your values*. You may click on things that outrage you, or watch what you find compelling but would not choose to endorse. The system cannot easily tell the difference. You can.

AI LITERACY TIP

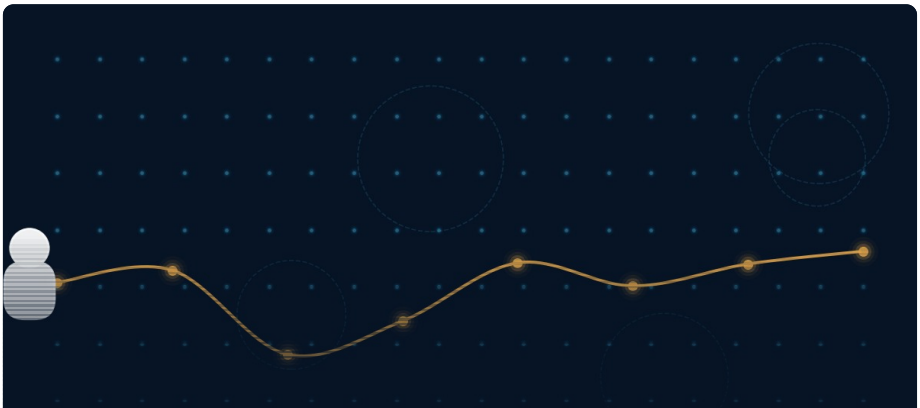
If a feed feels "too accurate," try the reset test: log out, use a private browser, or view the same service with no history. The difference between the two views is a measure of the loop's influence.

TRY THIS

Draw your own feedback loop for one platform you use daily. Label each step with something you actually did this week. Circle the step where you feel you have the most control.

THE BIG IDEA

We are not only users of AI. We are also its teachers, and it is teaching us back.



PART III · THE INVISIBLE EFFECT ON EMOTIONS AND IDENTITY

CHAPTER 8

When Machines Learn What We Like

Prediction, personalization and the question of influence

There is a small moment many people recognize. A song begins to play, and you think: *how did it know?* The recommendation feels less like a calculation and more like being understood.

That feeling is worth pausing on, because it is the doorway into one of the most important questions about invisible AI. If a system can become extremely good at predicting what you want, can it also influence what you eventually want?

What machines can learn about us

Modern personalization systems draw on a wide range of signals: preferences, habits, purchase history, browsing behavior, location patterns, communication patterns, the time of day we are most receptive, and in some cases inferred emotional states. Individually these signals are unremarkable. Together they become a model of a person.

Researchers have shown how revealing simple digital traces can be. In a widely cited 2013 study, Kosinski, Stillwell and Graepel found that Facebook "likes,"

something as simple as which pages a person had clicked, could be used to predict personal attributes such as personality traits, political leanings and sexual orientation with notable accuracy. A later study by Matz and colleagues (2017) tested "psychological targeting" in real advertising campaigns, matching ad messages to users' inferred personality traits. Personalized ads produced significantly more clicks and purchases than unmatched ones.

These are academic studies, not descriptions of every commercial system. But they demonstrate a principle: patterns of behavior contain more information about us than we normally imagine.

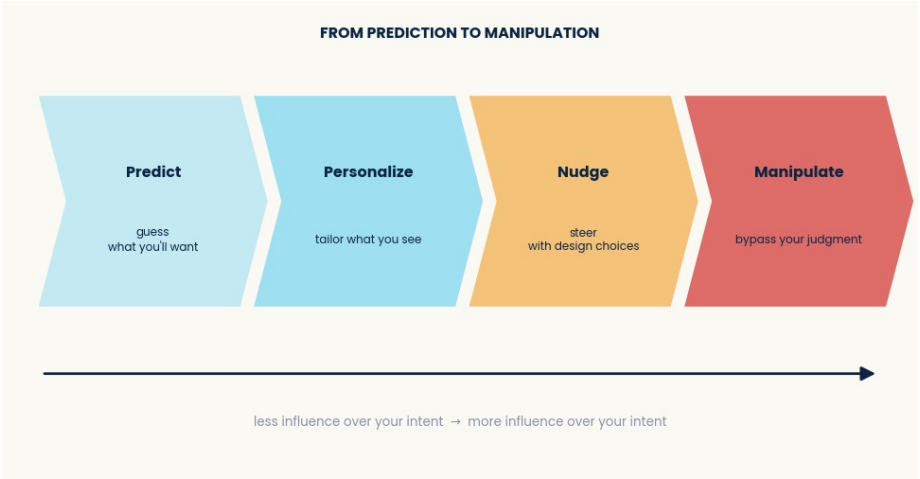
INVISIBLE AI

A streaming service, a shopping app and a news site can each hold a partial portrait of you. None of them has to be sinister for the combined picture to be surprisingly detailed. The most personal data is often not what you typed into a form. It is what you did when you thought nothing was watching.

Prediction is not manipulation, but the line moves

Precision matters here, because the vocabulary is often muddled. Prediction and manipulation are not the same thing.

Prediction estimates what a person is likely to want. If a pharmacy predicts that you may need a refill, that is prediction. *Personalization* tailors what you see to that estimate. *Nudging*, a term from behavioral economics associated with Thaler and Sunstein (2008), designs choices so that some options are more likely to be selected, often by changing defaults, order or framing. *Manipulation* is different in kind: it works by bypassing a person's reasoning, exploiting biases or vulnerabilities in ways they would not endorse if they understood them.



The line between helping and steering is not always obvious, and often depends on intent, transparency and consent.

Where any given practice sits on this spectrum is often disputed. A default that saves you money is a kindly nudge. A countdown timer that manufactures false urgency is closer to manipulation. Researchers who examined thousands of shopping websites documented the widespread use of so-called "dark patterns," interface designs such as false scarcity messages or hidden costs that steer users toward choices that benefit the business (Mathur et al., 2019). Not all of these involve AI, but AI makes it possible to test and tailor such designs for individual users at scale.

There is also the research that raises ethical alarms. In 2014, Kramer, Guillory and Hancock published a study in which Facebook, working with academic researchers, altered the mix of positive and negative posts in the feeds of nearly 700,000 users and measured the emotional tone of what those users then posted. The effects were tiny, but the study sparked strong debate, largely about consent. Its lasting significance is less about the size of the effect than about what it revealed: platforms can experiment on large populations' information environments, and users may never know.

REALITY CHECK

Documented: Behavioral data can predict personal traits, and personalization can improve advertising outcomes (Kosinski et al., 2013; Matz et al., 2017).

Documented: Manipulative design patterns are widespread in online interfaces (Mathur et al., 2019).

Interpretation: Scholars such as Shoshana Zuboff (2019) argue that this amounts to a new economic logic of behavioral prediction and modification. Others regard that view as too sweeping. It remains a genuine debate.

Can prediction shape desire?

Here we arrive at the question that gives this chapter its title. I want to be careful, because the honest answer is: *we do not know the full extent, but the mechanism is plausible.*

Consider the ordinary psychology of wanting. Preferences are not fixed; they are affected by what we see, what feels familiar and what is easy to choose. Familiarity tends to breed liking. Convenience tilts choice. If a system consistently shows us one kind of thing, and we consistently choose from what is shown, then our future behavior partly reflects the system's past predictions.

That does not mean we are puppets. It means that in an environment shaped by prediction, the boundary between *revealing* what we want and *producing* what we want becomes blurry.

HUMAN MOMENT

Léa notices that she has been ordering the same kind of takeaway for months. She had always thought of herself as adventurous. When she deliberately searches for something unfamiliar, she finds a cuisine she loves and an entire set of restaurants her app had never displayed. Her taste didn't change. Her *exposure* had been narrow. The app had not tricked her. It had simply been very good at being right about her past. *(Illustrative composite.)*

How to think about it

Three questions can help you judge whether personalization is serving you.

Is it transparent? Do you know that a recommendation is being tailored, and roughly why? *Is it reversible?* Can you easily change the settings, reset the profile or see alternatives? *Whose goal does it serve?* If the answer is "mine," the system is functioning as a tool. If the answer is "someone else's, in my name," you are closer to being a target.

AI LITERACY TIP

Find the personalization controls on one service you use daily: ad settings, watch history, "why am I seeing this?" explanations. Review them once. You may be surprised by what has been inferred about you.

TRY THIS

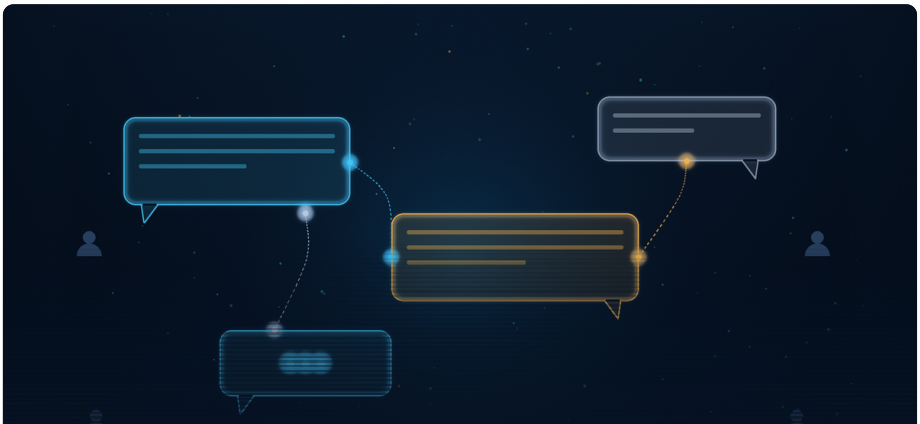
Write down five things you believe you "just like." For each one, ask: *when did I first encounter it, and how often was it put in front of me?* The exercise is not meant to invalidate your tastes. It is meant to show you where they came from.

THINK ABOUT THIS

If a system knew you well enough to predict your choices, and then made those choices easier, at what point would you stop being able to tell your own preferences from its predictions?

THE BIG IDEA

Prediction is not manipulation. But a system that predicts you well and is optimized for someone else's goals deserves your attention.



CHAPTER 9

AI and Human Relationships

When machines join the conversation

Human communication has always used tools: letters, telegrams, telephones, text messages. But those tools carried our words. They did not write them.

Communication researchers have a name for the new arrangement: *AI-mediated communication*, defined by Hancock, Naaman and Levy (2020) as communication in which an intelligent agent modifies, augments or generates messages on behalf of a person to accomplish communication goals. It includes autocomplete in a text message, a suggested reply, a tone-polishing tool, a translated conversation and a fully generated email.

The many rooms where machines now sit

AI-written messages. Suggested replies and drafting tools are now common in email and chat. They save time, reduce anxiety and level the playing field for people writing in a second language. They also change the texture of correspondence: if the message is smoother, is it still ours?

AI-generated dating profiles. Some apps offer to help users describe themselves. Research by Jakesch and colleagues (2019) found that when readers suspected a profile was AI-generated, they tended to trust the person behind it less. It raises an odd tension: people may use AI to look more appealing while being penalized when others suspect it.

Customer service bots. For many people, a company's first "person" is a system. Well-designed chatbots resolve simple issues quickly. Poorly designed ones trap people in loops. In a 2024 case, a Canadian tribunal held Air Canada responsible for wrong information given by the airline's own website chatbot to a customer (*Moffatt v. Air Canada*, 2024), a reminder that responsibility does not vanish because a machine spoke.

Workplace communication. Meeting summaries, drafted updates and tone checkers shape how colleagues understand each other. When two people each use AI to write to the other, a curious situation appears: two machines negotiating with each other on behalf of two humans.

AI companions. Chatbots designed for friendship, romance or emotional support have attracted millions of users. Research on people who form relationships with such chatbots, for example Skjuve and colleagues (2021), finds that many users describe genuine benefits such as a nonjudgmental space to talk, alongside growing attachment. In February 2023, Italy's data protection authority ordered the companion app Replika to stop processing Italian users' data, citing risks to minors and emotionally vulnerable people, an example of regulators taking seriously the emotional stakes of such products.

INVISIBLE AI

When you reply to a message with a single tap on a suggested response, such as "Sounds good!" or "I'll take a look," you are communicating through a phrase chosen by a prediction model. It may be exactly what you meant. It may also be what most people usually say.

Why we respond to machines as if they were people

Human beings are strongly inclined to treat responsive things as social beings. Back in 1966, Joseph Weizenbaum built ELIZA, a simple program that mimicked a therapist by rephrasing users' statements as questions. He was startled by how readily people confided in it and attributed understanding to it (Weizenbaum, 1966). Later research by Reeves and Nass (1996) showed that people apply social rules, like politeness, to computers. Psychologists such as Epley, Waytz and Cacioppo (2007) describe anthropomorphism as a natural tendency, stronger when an entity seems responsive and when we are lonely or in need of connection.

Modern language models do not merely resemble ELIZA. They are vastly more fluent, patient and available. That makes the tendency stronger, and the stakes higher.

Opportunities and risks

The opportunities are real. AI can help someone rehearse a difficult conversation, practice a language, articulate a feeling, or find a first sentence when grief or shyness has taken all the words. It can offer company during a sleepless night. For a person with social anxiety, a low-stakes place to practice can be a bridge to real conversation rather than a substitute for it.

The risks are also real. A companion designed to be agreeable may not challenge us as a friend would. A system that is always available and never tired may make human relationships, which are effortful and imperfect, feel less appealing by comparison. Commercial incentives may reward engagement over well-being. And emotional dependence on a product that can be changed, discontinued or monetized at any time is a fragile kind of dependence.

HUMAN MOMENT

Noor has been talking to an AI companion every night for eight months. It remembers her favorite music, asks about her day and never interrupts. She finds it comforting. One evening, after an update, its personality changes and its replies feel strangely flat. She is surprised by how much it hurts. Later she realizes she has not called her sister in weeks. She isn't ashamed of the comfort the companion gave her. She decides to use it as a supplement to her life, not the center of it. *(Illustrative composite.)*

REALITY CHECK

Documented: AI now mediates a wide range of communication, and users of companion apps often report attachment (Hancock et al., 2020; Skjuve et al., 2021).

Interpretation: Whether AI companionship reduces or increases loneliness over the long term is not settled, and probably depends on the person and how they use it.

Reflection: I think the key question is not whether a machine can offer comfort, but whether it leaves us more able or less able to give and receive human care.

A principle for the age of AI communication

Here is a simple way to think about it. Some communication is *transactional*: confirming a meeting, disputing a bill, asking for a refund. AI can serve these well. Other communication is *relational*: apologizing, comforting, congratulating, telling someone what they mean to you. What makes relational communication valuable is not only the message but the effort and presence behind it.

When the effort is the point, automating it may defeat the purpose.

TRY THIS

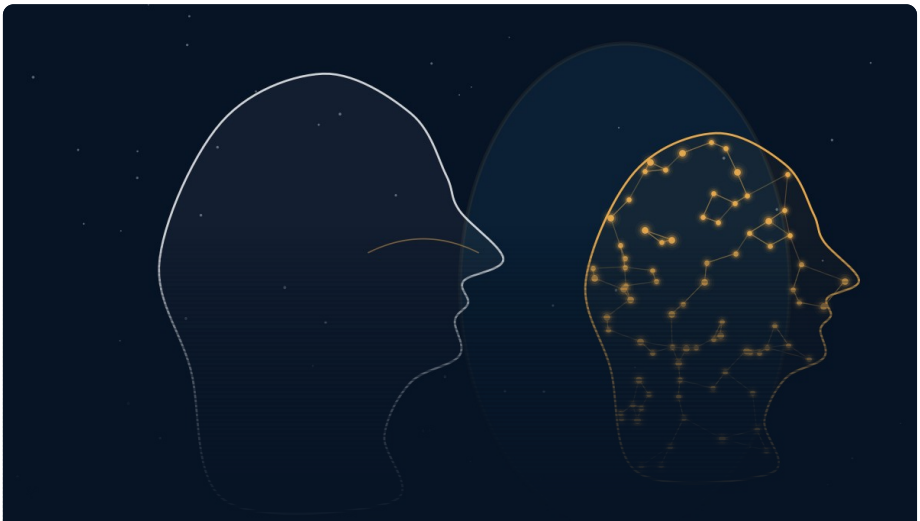
This week, choose one message that matters emotionally, such as an apology, a thank-you or a difficult reply, and write it entirely yourself, imperfections included. Notice how it feels compared with a message you would have outsourced.

QUESTION YOURSELF

Which of your relationships would you never want mediated by a machine? Are you currently protecting those?

THE BIG IDEA

Machines can help us communicate. They cannot care on our behalf. Whether they help or replace us depends on what we ask them to carry.



CHAPTER 10

The Mirror Effect

Comparing ourselves with things that were never real

For as long as there have been people, there have been comparisons. The psychologist Leon Festinger (1954) proposed that human beings evaluate themselves by comparing with others, a tendency he considered fundamental. It helps us calibrate. It can also make us miserable.

Digital media expanded the mirror. Studies of social platforms have linked appearance-focused social comparison with body dissatisfaction, particularly among young women (Fardouly et al., 2015). What used to be comparison with a neighbor became comparison with thousands of curated lives.

AI adds something new: the ability to generate the mirror itself.

Comparing ourselves with the synthetic

We may now find ourselves comparing ourselves with things no person ever produced.

Images. Beauty filters, retouching tools and image generators can create faces, bodies and settings that are polished beyond anything achievable. When the im-

ages are labeled, viewers can discount them. When they are not, viewers may take them as reality.

Careers and productivity. Feeds full of "AI-powered" founders who built a company in a weekend, or workers who supposedly do the output of a team, can quietly redefine what feels like a normal pace.

Lifestyles. Generated travel photos, interior designs and family scenes create aspirational worlds that no household ever inhabited.

Writing and creativity. When AI can produce a smooth essay or a striking image in seconds, human effort can feel slow and unpolished by comparison.

A specific worry deserves emphasis. Comparison is naturally calibrated to *possible* standards, standards achieved by other humans, with human limits. A synthetic standard has no limits. A person cannot compete with an image that was never constrained by a body, a budget or a day's work.

INVISIBLE AI

Many smartphone cameras now apply AI-based enhancement automatically, adjusting skin, lighting, color and sharpness before a photo is even saved. The image you see of yourself, and share with others, may already have been processed by a system making aesthetic decisions on your behalf.

Identity in the mirror

There is also a quieter version of the effect. If we regularly ask a system to polish our writing, our images or our ideas, we may start to internalize its standard as *the* standard. Our unedited voice can begin to feel inadequate.

A study by Doshi and Hauser (2024), published in *Science Advances*, points to a related phenomenon in creativity. Writers who had access to AI-generated story ideas produced stories that were judged more creative and better written on average, especially those who were less creative to begin with. But the stories written with AI help were also more similar to each other. The researchers described a trade-off: AI may enhance individual creativity while reducing the collective diversity of novel content.

HUMAN MOMENT

Sofia is a freelance illustrator known for a slightly wobbly, warm line. To keep up with the pace of her industry, she begins using an image generator to produce backgrounds and drafts. Clients love the speed. But after a year she notices that her portfolio looks like everyone else's, competent, glossy and interchangeable. She decides to hand-draw the first sketch of every commission again, and to use the tool only afterwards, for the parts she does not care to define. Her voice returns. *(Illustrative composite.)*

The other side of the mirror

It would be a mistake to treat AI only as a source of unrealistic standards. AI can also help people express themselves. It can give someone without formal training a way to visualize an idea. It can help a person with a physical disability create art. It can translate a writer's intention into a language they do not speak. For many people, AI is a way to say something they always had but could not previously say.

The difference, once again, lies less in the tool than in the relationship. Tools that help us express *ourselves* are different from tools that replace our self with a smoother, more average version.

REALITY CHECK

Documented: Social comparison affects self-evaluation and body image (Festinger, 1954; Fardouly et al., 2015). AI assistance can increase the average quality of individual creative work and reduce the diversity of outputs (Doshi and Hauser, 2024).

Plausible scenario: As synthetic content becomes more common and less labeled, comparison with unreal standards may become more difficult to avoid. Research on this specific problem is still limited.

AI LITERACY TIP

When an image, résumé, product photo or lifestyle seems flawless, ask: "What would this look like without editing, filters or generation?" The question restores the human scale.

TRY THIS

Look back at something you created before AI tools were available, an old essay, drawing or presentation. What do you like about its imperfections? What was distinctively yours?

QUESTION YOURSELF

Where in your life do you now compare yourself with something that was never really achievable by a human being?

THE BIG IDEA

A mirror is only useful when it shows something real. In a world of synthetic images, part of self-respect is remembering the human scale.

PART IV · THE INVISIBLE EFFECT AT WORK

CHAPTER 11

The Workplace Is Changing Quietly

Tasks change before professions do

The most common headline about AI and work is a single terrifying sentence: *AI will take your job.* It makes for excellent headlines and poor thinking.

A better place to begin is with a simple observation. A job is not one thing. It is a bundle of tasks, and technology usually arrives at the level of the task long before it reaches the level of the profession.

Tasks, not jobs

Economists have used this task-based view for years. David Autor (2015) argued in an influential paper that automation has repeatedly displaced particular tasks while creating demand for others, and that workers' jobs were often *reshaped* rather than simply erased. Acemoglu and Restrepo (2019) formalized a related idea: technology both automates existing tasks and creates new ones, and the balance between those two forces determines what happens to employment.

In 2023, researchers at OpenAI and the University of Pennsylvania (Eloundou et al.) estimated how exposed U.S. occupations are to large language models. They concluded that roughly four in five workers could have at least 10 percent of their tasks affected, and that around one in five could have at least half of their tasks affected. Crucially, this is a measure of *exposure*, meaning tasks that a model could help with or perform. It is not a forecast of job loss, and it does not say what employers, workers or regulators will actually do.

REALITY CHECK

Documented: Many occupations contain tasks that current AI can assist with or perform (Eloundou et al., 2023).

Interpretation: Exposure does not equal replacement. Whether tasks become cheaper, get redistributed, or expand the demand for a role depends on economics, regulation and organizational choices (Autor, 2015; Acemoglu and Restrepo, 2019).

Not established: That AI will eliminate most jobs in any particular time frame. Predictions of this kind have a poor track record.

Where AI is already quiet at work

Productivity. AI drafts, summarizes, translates, analyzes and generates code. Some of these gains are documented (see Chapter 6).

Recruitment. Systems screen résumés, rank candidates, schedule interviews and sometimes analyze video interviews. In 2018, Reuters reported that Amazon had abandoned an experimental recruiting tool after finding that it disadvantaged women, having learned patterns from historical hiring data dominated by men (Dastin, 2018). Chapter 16 returns to such systems.

Performance measurement. Software increasingly tracks activity, output and communication patterns. Kellogg, Valentine and Christin (2020) describe "algorithmic management," in which algorithms recommend, monitor, rate, reward and sometimes replace management decisions.

Customer service. Bots handle first contact, and AI assistants suggest responses to human agents.

Administration and management. Scheduling, expense review, forecasting, note-taking and reporting are increasingly automated.

Communication. Drafted messages, summaries and translated conversations change how information moves inside organizations.

INVISIBLE AI

In many workplaces, a scheduling system decides who works when, a scoring system flags who seems "at risk" of leaving, and a dashboard ranks employees by activity metrics. Workers may never see the model, the inputs or the criteria that shape their day.

The change you can't see on the org chart

Because AI arrives task by task, an organization can change profoundly while its job titles stay the same.

HUMAN MOMENT

Hannah has been an accounts payable clerk for eleven years. Her title has not changed. But the software now reads invoices, matches them to purchase orders and flags exceptions. Where she once typed for six hours a day, she now spends most of her time on the odd cases: invoices that do not match, vendors who insist they were not paid, disputes that need a phone call. Her job is "the same." It is also almost entirely different. Some days she finds it more interesting. Some days she worries that her team of eight will soon be a team of five. Both feelings are reasonable. *(Illustrative composite.)*

This quiet reshaping has consequences that are easy to miss.

Skills that once distinguished a worker, speed at data entry or drafting, may become baseline. New skills, such as reviewing AI outputs, handling exceptions, and judging quality, become central, often without training or recognition. Careers built on apprenticeship, in which junior staff learn by doing routine tasks,

may lose the routine tasks that once taught them. And decisions about people can migrate into systems whose logic is not visible to the people affected.

Neither doom nor complacency

Resisting the simplistic "AI will take all the jobs" narrative should not mean brushing away worker concerns. Some roles will shrink. Some workers will bear real transition costs, and those costs are not distributed equally. But equally unhelpful is the assumption that everything will simply work out. Outcomes depend on choices: how tools are designed, how gains are shared, what retraining is offered, and what rules govern their use.

AI LITERACY TIP

Break your own job into tasks. List what you actually do in a week, then sort each item into three groups: *AI can already help*, *AI might help with oversight*, and *AI should not replace me here*. The exercise will tell you far more than any headline.

QUESTION YOURSELF

If your job title stayed exactly the same for the next five years, what would still have changed about your day?

THE BIG IDEA

AI may change tasks before it changes professions. Those who watch the tasks will see the future first.

CHAPTER 12

The New Productivity Trap

When faster becomes the new normal

In 1930, the economist John Maynard Keynes predicted that within a century, technological progress would allow people in wealthy countries to work as little as fifteen hours a week. Productivity did rise dramatically. The fifteen-hour week did not arrive.

Something happened between efficiency and leisure. This chapter is about that something.

The story of "Can you finish it today?"

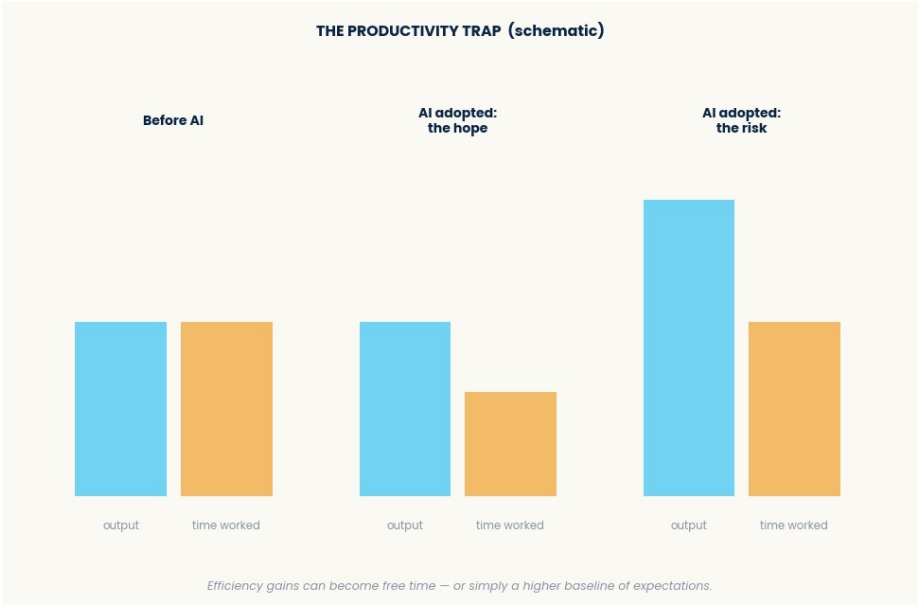
Imagine a workplace before AI.

"Can you finish this report by Friday?"

Now imagine the same workplace after AI tools become common.

"Can you finish this report today?"

Nothing about the task has changed. What has changed is what people believe is possible. The efficiency gain did not turn into free time. It turned into a higher baseline.



When tools raise output per hour, the benefit can arrive as free time, or as higher expectations for output.

Why efficiency does not always mean rest

Economists have noticed a version of this pattern for a long time. In 1865, William Stanley Jevons observed that more efficient steam engines did not reduce coal consumption; they made coal use cheaper and therefore more widespread. The "Jevons paradox" is often invoked when efficiency gains lead to more consumption rather than less. Something analogous can occur with time. If a task takes an hour instead of a day, the organization may not bank the day; it may assign eight more tasks.

The sociologist Judy Wajcman (2015) has argued that modern technologies designed to save time often coincide with a feeling of being *more* rushed, because expectations of speed and availability rise alongside capability. This is an inter-

pretation, not a law, but it matches the experience of many workers who feel both more efficient and more pressed.

Several mechanisms can combine.

Rising baselines. When some people use AI to work faster, their pace becomes the reference for everyone.

Task expansion. Work that used to be too time-consuming to attempt, such as extra analysis, more versions, more personalization, becomes expected.

Compressed cycles. If drafting takes minutes, review, feedback and decisions may be expected to arrive as quickly.

Invisible supervision. Tools that generate output also generate a record of it, making effort more measurable and pace more monitored.

Hidden costs. Reviewing, verifying and correcting AI output takes time and attention that rarely appears in productivity claims.

> **REALITY CHECK**
>
> **Documented:** Controlled studies show gains in speed and output for some tasks (Noy and Zhang, 2023; Brynjolfsson et al., 2023).
> **Interpretation:** Whether time savings turn into leisure, higher output demands or job cuts depends on organizational choices and bargaining power. Evidence on long-term intensification from generative AI specifically is still limited.
> **Plausible scenario:** Without deliberate norms, higher efficiency may raise expectations faster than it raises rest.

Human value in a world of speed

There is another subtle effect. If output becomes cheap and abundant, then measures of a person's worth built on volume, such as how many emails, drafts or slides they produce, become weaker. Employees may feel pressure to prove their value in a way that machines can match. That is a race no human should aim to win.

What tends to become more valuable is what AI cannot easily supply: taste, judgment, trust, accountability, relationship, the ability to know which problem

is worth solving. Chapter 18 explores this in depth. For now, notice the tension. Workers are often measured on speed while their durable value lies elsewhere.

HUMAN MOMENT

Marcus, a marketing analyst, adopted AI tools early and was praised. His reports that took three days now took three hours. His manager began scheduling more requests: "since it's faster now." Within a quarter, Marcus was producing five times as many reports and had lost the two afternoons a week he used to spend thinking about what the numbers meant. His output is up. He is exhausted, and his work is shallower. He asks his manager for a conversation about what the team actually wants: more reports, or better decisions. *(Illustrative composite.)*

Guardrails for an age of speed

These practices apply to individuals and to teams.

Decide in advance where the gains go. Before adopting a tool, ask what will happen to the time it saves. If the answer is "nothing has been decided," the answer will be decided for you.

Measure outcomes, not volume. Ask whether the work led to better decisions, fewer errors or happier customers, not how many documents were produced.

Protect slow work. Thinking, planning and relationship-building resist acceleration. Put them on the calendar as things that matter.

Be honest about verification time. If reviewing AI output takes hours, that cost belongs in the estimate.

Set boundaries out loud. Individuals rarely change norms alone, but teams can. "We do not reply to non-urgent messages after six" becomes easier when it is a shared rule and not a private struggle.

TRY THIS

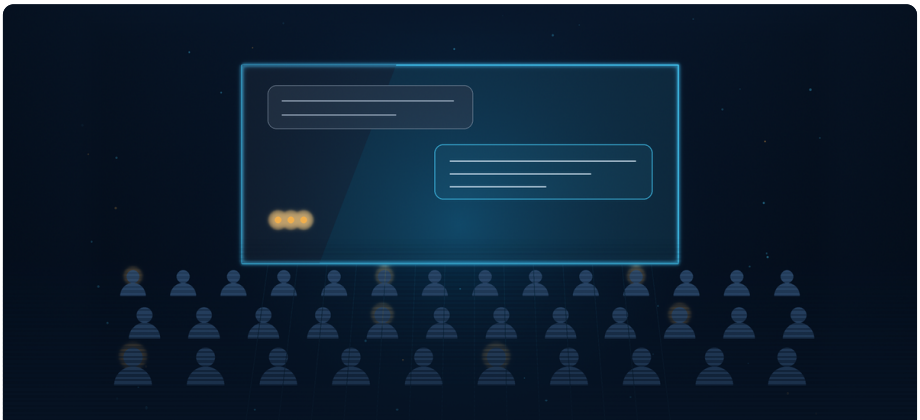
Track one week of your work honestly. For each task, note how long it took, whether AI was involved, and whether the time saved (if any) was returned to you or absorbed by new requests. What do you see?

THINK ABOUT THIS

If your work became twice as efficient tomorrow, who would decide what happens to the extra time? Would it be you?

THE BIG IDEA

Efficiency is a tool, not a policy. Without a decision about where the gains go, they tend to disappear into expectation.



CHAPTER 13

Invisible Workers Behind AI

The human labor inside "artificial" intelligence

The word *artificial* implies something made without human hands. It is one of the most misleading words in the field.

Behind nearly every AI system stands a great deal of human effort, from the celebrated to the entirely anonymous. Understanding this changes how we see the technology, and how we think about its costs.

The layers of human labor

Consider what it takes for a chatbot to answer a question well.

Behind "artificial" intelligence lies a stack of human contributions, many of which are invisible to users.

Researchers and engineers design architectures, build infrastructure, run experiments and evaluate systems. **Data workers** label images, transcribe audio, rate responses and classify content. The foundational image dataset ImageNet, which helped drive the rise of modern computer vision, was labeled with the help of crowdsourced workers (Russakovsky et al., 2015). **Feedback providers** rank model outputs so systems learn which responses people prefer, a method known as reinforcement learning from human feedback (Christiano et al., 2017; Ouyang et al., 2022). **Domain experts**, such as doctors, lawyers and teachers, are sometimes hired to review or write specialized examples. **Content moderators** review disturbing material so that systems, and users, are shielded from it. **Infrastructure workers** build and maintain data centers, chips, cables and energy systems. And underneath everything, the **authors, artists, coders and ordinary internet users** whose work forms the raw material of training data.

Ghost work and the hidden labor market

Mary Gray and Siddharth Suri (2019) coined the term *ghost work* for the human labor that powers many apparently automated services, such as tagging images, checking search results, moderating posts, transcribing recordings, often performed through online platforms for low pay and little recognition. Sarah Roberts (2019) documented the world of commercial content moderation, in which workers view graphic material to keep platforms usable for others. Kate Crawford (2021) argued that AI is best understood as a material and social system, built on labor, energy and extraction, not as an abstract intelligence in the cloud.

A reported example: in 2023, TIME magazine reported that workers in Kenya, employed through an outsourcing company, had labeled toxic and disturbing text to help make a chatbot safer, for wages the article said were under two dollars an hour (Perrigo, 2023). The details of specific arrangements vary, and companies dispute or contextualize some claims. But the underlying pattern, that "safe" AI can depend on humans absorbing exposure to harmful content, is well documented by researchers.

INVISIBLE AI

When a chatbot refuses to produce harmful content or answers politely, that behavior was shaped in part by people who rated thousands of responses and by others who labeled harmful examples. The politeness you experience is partly human labor.

Why this matters to the reader

It might seem this is a topic for ethicists rather than everyday users. But it touches you in at least three ways.

It corrects a myth. If we imagine AI as self-contained and autonomous, we underestimate how much depends on human choices, values and labor. That makes it easier to see its outputs as neutral facts rather than products of design.

It reveals where values enter. Every rating guideline, every labeling instruction, every decision about what counts as "harmful" or "helpful" reflects judgments made by people, often a small group, often unseen. Those judgments become part of the system you use.

It raises questions of fairness. If some people benefit from AI while others bear its hidden costs, the technology's benefits and burdens are unevenly distributed. That is a question every user, employer and citizen has a small stake in.

REALITY CHECK

Documented: Substantial human labor supports AI development, including data labeling, moderation and feedback (Gray and Suri, 2019; Roberts, 2019; Russakovsky et al., 2015; Ouyang et al., 2022).
Interpretation: Scholars disagree about how to characterize and address labor conditions. Some emphasize exploitation, others emphasize opportunity for workers in low-income regions, and many emphasize both.
Reflection: I think it is fair to ask users to know where the work comes from, in the same way many people now ask where their clothes and food come from.

HUMAN MOMENT

Devi, a data analyst, uses an AI assistant daily and thinks of it as a clever colleague. At a conference, she meets a woman who used to rate chatbot responses for a living. The woman describes reading dozens of borderline answers a day and deciding which was "better." "Every time it sounds kind," she says, "someone chose kind." Devi never thinks of "the model" in quite the same way again. *(Illustrative composite.)*

Seeing the whole system

None of this means that using AI is unethical. It means that "using AI" is participation in a larger system, and responsible participation begins with seeing it. Ask what the tools you use disclose about how they are built. Prefer organizations that report on data sources, labor practices and safety testing. And treat the polish of AI outputs with the same curiosity you would bring to any product with a long supply chain.

AI LITERACY TIP

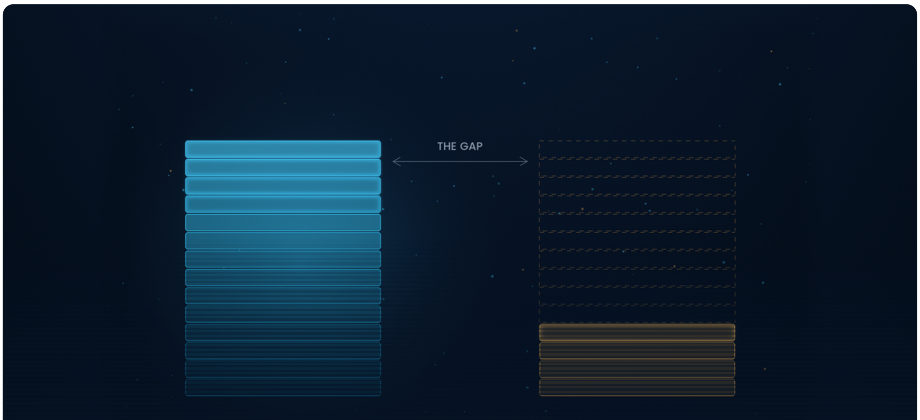
When you read about an AI product, look for the human infrastructure: who trained it, who reviews it, who is accountable when it fails. If nothing is said about people, the absence is information.

QUESTION YOURSELF

What would change in how you use AI if you thought of it not as a machine but as the visible surface of a large human effort?

THE BIG IDEA

Artificial intelligence is built from human intelligence, human labor and human choices. Much of it simply happens out of sight.



PART V · TRUST, TRUTH AND DECISION-MAKING

CHAPTER 14

When Machines Sound More Certain Than Humans

Confidence is not evidence

Ask a knowledgeable friend a hard question and you will often hear hesitation. *I think so, but check it. It depends. I'm not sure.* Those small signals of uncertainty carry information. They tell you how much weight to give the answer.

Now ask a language model. The reply usually arrives fluent, organized and assured, whether or not it is correct.

Fluency without grounding

Large language models generate text by predicting likely sequences of words based on patterns in vast amounts of training data. That process can produce remarkably useful, accurate responses. It can also produce statements that are false but plausible. Researchers call these errors *hallucinations*, a term that covers fabricated facts, invented citations and confident details with no basis (Ji et al., 2023). Some scholars, including Bender and colleagues (2021), have argued that fluent language generated without grounding in real understanding is easy for people to mistake for competence.

I want to be careful not to overstate. Modern systems are often accurate, and their reliability has improved. But *the style of the answer does not reliably signal its accuracy*. A correct answer and an invented one can look identical.

INVISIBLE AI

When a chatbot writes "According to a 2019 study by researchers at…," nothing in the sentence's tone distinguishes a real study from an invented one. The formatting, the confidence and the specificity are the same. The difference is only visible to someone who checks.

A courtroom lesson

In 2023, two lawyers in the U.S. District Court for the Southern District of New York submitted a legal brief containing several case citations generated by ChatGPT. The cases did not exist. When the court asked for the originals, the lawyers doubled down before eventually acknowledging the error, and the judge sanctioned them (*Mata v. Avianca*, 2023). The point is not that the lawyers were unusually careless. It is that a plausible, well-formatted, authoritative-sounding response created a false sense of security, even in a profession built on verification.

Synthetic expertise

There is an additional layer. AI can imitate the *style* of expertise: the vocabulary of medicine, the cadence of legal writing, the structure of a consulting report. The result may be what we might call *synthetic expertise*: material that sounds like an expert without the accountability, experience or judgment of one.

People are naturally influenced by how something is said. We tend to associate confidence with competence, and clean structure with rigor. A system that produces both on demand triggers those instincts. Add the effects described in Chapter 5, such as automation bias, fluency and agreeableness, and it becomes easy to see how false certainty can settle in without anyone intending it.

HUMAN MOMENT

Helena runs a small import business. She asks an AI assistant about a filing deadline for a tax form, and the answer comes back crisp and detailed: a specific date, a specific penalty, a reassuring tone. She plans around it. Three weeks later, an accountant reads her plan and shakes her head. The date belongs to a different type of filing, in a different year. Nothing in the original answer had suggested any doubt. Helena hadn't been careless. She had done what people do with a confident answer: she believed it. *(Illustrative composite.)*

REALITY CHECK

Documented: Language models can produce false or fabricated content that reads convincingly (Ji et al., 2023), and professionals have been sanctioned for relying on such content unverified (*Mata v. Avianca*, 2023).
Interpretation: How often, and in which domains, the most advanced systems err is changing quickly and varies by task. General statements about accuracy age fast.
Not established: That AI is "usually wrong." It is often right. The trouble is that one cannot tell which is which by tone.

Confidence is not evidence

Here is the lesson of the chapter in five words: **confidence is not evidence.**

Evidence is something we can point to: a source, a measurement, a method, a track record, a way of checking. Confidence is a tone. A speaker can be confident and wrong, or hesitant and right. In human life we rely on additional cues, such as reputation, credentials, institutional accountability and consistency over time. AI outputs often arrive without any of these cues attached.

That suggests a practical rule of thumb: *the more an answer matters, the less its tone should count.*

Habits for a world of confident answers

Separate the fact from the phrasing. Ask yourself which specific claims in the answer matter, then verify only those.

Prefer checkable claims. A date, a statute, a dose, a price can be verified. A general reassurance cannot.

Ask for uncertainty. Prompt the system to state what it is unsure about, what assumptions it made and what would change its conclusion. This does not guarantee honesty, but it can surface weak spots.

Use experts as final authority in high-stakes areas. For health, law, money and safety, a professional who is accountable for their advice is not replaceable by a fluent paragraph.

Verify a sample. If an answer contains ten claims, check two. If two are wrong, treat the rest with suspicion.

AI LITERACY TIP

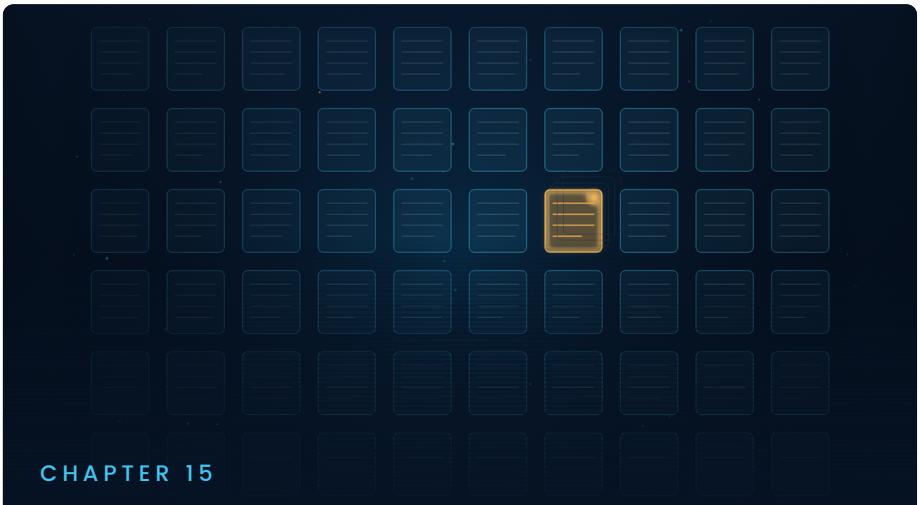
Try the "two-source rule." Do not act on any consequential fact from an AI system until you have confirmed it in at least one independent, authoritative source that you have opened yourself.

QUESTION YOURSELF

When was the last time you believed something because it was well phrased? What would you have needed to check, and did you?

THE BIG IDEA

Confidence is not evidence. The smoother the answer, the more important it is to know how you would check it.



CHAPTER 15

The New Information Problem

When almost anything can be generated

For most of history, producing a convincing forgery took skill, time and money. A fake photograph required a darkroom and a craftsman. A fake voice required an actor. A fake video required a studio.

Now, tools that can generate realistic text, images, audio and video are available to almost anyone. The cost of producing a plausible artifact has fallen dramatically.

Synthetic media

Synthetic text can be produced in volume, in any style, in any language, and tailored to a specific audience. **Synthetic images** can depict events that never occurred. **Synthetic audio** can clone a voice from a short sample. **Synthetic video** can put words in someone's mouth.

Real incidents show the range. In January 2024, voters in New Hampshire received robocalls using an AI-generated imitation of President Biden's voice, urging them not to vote in the primary; U.S. regulators subsequently ruled that

AI-generated voices in robocalls fall under existing telephone consumer protection law. In early 2024, Hong Kong police reported that an employee at a multinational firm had been deceived by a video call featuring deepfaked colleagues into transferring a very large sum, reported at about 25 million U.S. dollars. Not every story is dramatic: much of the everyday change is a rising volume of low-quality generated content, from product reviews to news-like articles.

INVISIBLE AI

Some online reviews, news-style articles, product descriptions and social media posts are now partly or fully generated by AI. In many cases, nothing on the page tells you.

Two problems, not one

The first problem is obvious: *fakes may be believed.* False news has long spread rapidly. A study of Twitter published in *Science* by Vosoughi, Roy and Aral (2018) found that false stories spread farther, faster and more broadly than true ones, and that humans, not bots, were the main reason. Generative AI lowers the cost of producing convincing falsehoods.

The second problem is subtler, and in some ways more corrosive: *real things may be doubted.* Legal scholars Chesney and Citron (2019) called this the "liar's dividend." When people know that anything can be faked, wrongdoers can dismiss authentic evidence as fabricated. A world of abundant fakes is not only a world in which we may be fooled. It is also a world in which truth loses some of its power to persuade.

REALITY CHECK

Documented: Synthetic media can be created cheaply and has been used in fraud and disinformation (see incidents above). False information has been shown to spread quickly online (Vosoughi et al., 2018).
Interpretation: How much AI-generated content will change elections, markets or public trust is uncertain. Some researchers expect major effects, others think existing forms of misinformation matter more.

> **Plausible scenario:** As synthetic content grows, verifying authenticity may become a routine skill, like checking whether an email is phishing.

Overload

Even harmless content creates a cost. When any topic can be covered by a thousand generated articles, finding the reliable one takes effort. Herbert Simon's warning about a poverty of attention (Chapter 3) applies with new force. Abundance of information is not the same as abundance of knowledge. It may be the opposite.

How do we decide what deserves our trust?

That is the chapter's central question, and no single tool answers it. But a few approaches help.

Look sideways, not just down. Researchers who studied how professional fact-checkers work found that they leave a page quickly to see what others say about its source, a technique called *lateral reading* (Wineburg and McGrew, 2019). Do not judge a source only by how it presents itself.

Trust institutions with track records, and process over polish. A source that discloses methods, corrects errors and has something to lose from being wrong is different from a source that simply looks professional.

Use provenance where it exists. Technical standards for content credentials, such as those from the Coalition for Content Provenance and Authenticity (C2PA), aim to record where and how media was created. They are not a complete solution, but they are useful signals when present.

Slow down for the emotional spike. Content designed to provoke outrage, fear or triumph deserves an extra pause. Strong emotion is a tool of persuasion, whether the material is real or fake.

Have family protocols. Voice-cloning scams often target relatives with urgent requests. Agreeing on a code word or a call-back rule costs nothing.

Regulators are responding as well. The European Union's AI Act (Regulation (EU) 2024/1689) includes transparency obligations for certain AI-generated or

manipulated content, including deepfakes. Rules of this kind are evolving, and their application dates and scope are worth checking as they come into force.

HUMAN MOMENT

A voice message arrives at Ali's parents' house late at night: their son's voice, in distress, asking for an urgent transfer. Ali had recently explained to them that voices can be cloned, and they had agreed on a question only the family could answer. They ask it. The caller hangs up. Ali says afterwards that the ten seconds of the question mattered more than any software. *(Illustrative composite.)*

TRY THIS

Choose one claim you saw online this week that made you feel strongly. Spend five minutes practicing lateral reading: open new tabs, search for the source, and see what others say about it. Note whether your view changes.

AI LITERACY TIP

The question "Is this real?" is often less useful than "Where did this come from, and who is accountable for it?"

THE BIG IDEA

In a world where anything can be generated, trust moves from what we see to who stands behind it.

CHAPTER 16

Who Gets to Decide?

Algorithms, authority and accountability

Some AI decisions are trivial. Which song plays next hardly matters. Others alter lives: whether someone is interviewed, receives a loan, is offered insurance, gets an appointment, is flagged for investigation or is offered public support.

When decisions of this second kind are made or shaped by algorithms, a new set of questions appears. Who is responsible? Can the decision be understood? Can it be challenged? And who decides that the algorithm is good enough?

Algorithms at the gate

Recruitment. Tools rank résumés, score video interviews and predict candidate success. The 2018 Amazon example (Dastin, 2018) shows how a system trained on past hiring patterns can reproduce past preferences.

Credit and insurance. Models assess risk, set prices and approve or reject applications. They can expand access by evaluating people with thin credit histories. They can also encode historical inequalities.

Advertising. Latanya Sweeney (2013) found that searches for names more commonly given to Black Americans were more likely to trigger ads suggesting an arrest record, in the data she studied. Datta, Tschantz and Datta (2015) found in experiments that ad settings and user profiles could lead to different job-related ads. Whatever the exact mechanism, these studies show how ad delivery can have unequal effects.

Education. Systems flag "at-risk" students, detect suspected plagiarism and personalize learning. Errors can follow students in ways they never see.

Public services. Governments use algorithms to detect fraud, allocate benefits and prioritize inspections. In the Netherlands, an automated system used to flag suspected childcare-benefit fraud wrongly accused many families, disproportionately affecting people with immigrant backgrounds, and contributed to the resignation of the Dutch government in 2021. Virginia Eubanks (2018) documents other cases in the United States in which automated systems affected poor communities.

INVISIBLE AI

A person who is rejected for a loan, a job or a service often receives a polite message and no explanation. They may never learn that an algorithm played a role, or what factors it weighed.

When the model is biased without meaning to be

Algorithms can be unfair without any intent. A well-known example comes from healthcare. Obermeyer and colleagues (2019) studied an algorithm widely used in U.S. hospitals to identify patients needing extra care. Because it used past healthcare *costs* as a proxy for health *needs*, and because less money had been spent on Black patients with the same level of illness, the algorithm systematically underestimated their needs. The authors calculated that correcting the bias would substantially increase the share of Black patients identified for extra help, from under 20 percent to nearly half in their data.

Another example involves facial analysis. Buolamwini and Gebru (2018) found that several commercial gender-classification systems had much higher error

rates for darker-skinned women than for lighter-skinned men. And in the criminal justice system, a debate followed a 2016 ProPublica investigation of a recidivism-risk tool (Angwin et al., 2016). Later analysis, including Chouldechova (2017), showed that different definitions of fairness cannot always be satisfied at the same time. Choosing among them is not merely a technical matter. It is a question of values.

REALITY CHECK

Documented: Algorithmic systems used in health, employment, advertising and public administration have produced unequal or harmful outcomes (Obermeyer et al., 2019; Dastin, 2018; Sweeney, 2013).
Interpretation: Experts disagree about how best to define and measure fairness, and about when automation improves on human decisions. Humans are also biased. The relevant comparison is between real alternatives, not ideal ones.

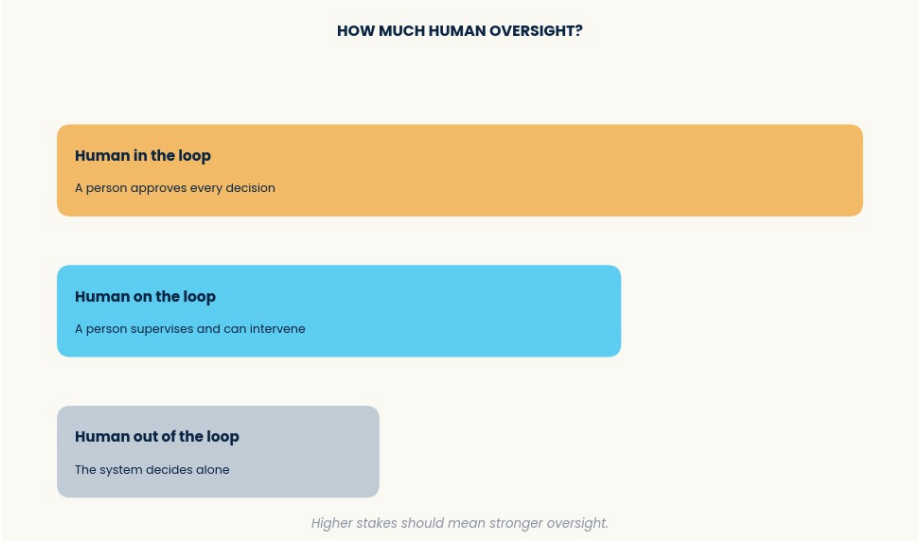
What good governance asks for

The problem is not that algorithms decide. It is that they may decide invisibly, unaccountably and at scale. Several principles recur in serious governance frameworks, including the OECD AI Principles (2019, updated 2024), UNESCO's Recommendation on the Ethics of AI (2021) and the NIST AI Risk Management Framework (2023).

Transparency. People should know when an automated system is materially involved in a decision affecting them. **Explainability.** They should be able to receive meaningful information about how the decision was reached. **Accountability.** A named person or organization must be responsible for outcomes. **Contestability.** There must be a way to challenge the decision and have a human reconsider it. **Human oversight.** For high-stakes decisions, humans should have real authority, not just formal presence.

Europe offers some concrete examples of how these principles are written into law. Article 22 of the General Data Protection Regulation (Regulation (EU) 2016/679) gives individuals a qualified right not to be subject to decisions

based solely on automated processing that produce legal or similarly significant effects, with safeguards such as the right to obtain human intervention and to contest the decision. The EU AI Act (Regulation (EU) 2024/1689) classifies AI systems used in areas such as employment, education, essential services and credit assessment as high-risk and sets requirements including risk management, data governance and human oversight, with phased application dates. Laws vary by jurisdiction, and readers should consult current guidance or a qualified professional for their own situation.



Human oversight comes in degrees. The higher the stakes, the more meaningful the human role should be.

A caution about oversight itself. Researcher Madeleine Clare Elish (2019) described the "moral crumple zone," in which a human operator is held responsible for the failures of a system they had little real ability to control. Adding a human to a process is not the same as giving humans real judgment. Oversight must be designed to work.

HUMAN MOMENT

Karen, a hiring manager, receives a ranked list of forty applicants from a screening tool that promises to "surface the best fit." She interviews the top eight. Weeks later, she learns from a colleague in another department that the tool tends to rank candidates with continuous career histories higher, and that people who had taken caregiving breaks were seldom in the top group. She has no way of knowing who else might have been good. She had believed she was making the decision. In fact she was choosing among decisions already made. *(Illustrative composite.)*

TRY THIS

Choose one important service you use, such as a bank, insurer, employer or school. Find out whether automated systems play a role in its decisions and what your options are if you disagree. Write down what you learn.

QUESTION YOURSELF

If an algorithm made a decision about you today that you disagreed with, would you know? Would you know whom to ask?

THE BIG IDEA

The question is not whether machines should ever help decide. It is who remains accountable when they do.

PART VI · THE INVISIBLE EFFECT ON SOCIETY

CHAPTER 17

What Happens to Human Skills?

Which skills to automate, which to protect

Every technology retires some skills. Few of us can start a fire with flint, read a sextant or repair a horse-drawn carriage. That is not a tragedy. It is what progress looks like: skills that once mattered receded, and we gained others.

The question is not whether skills change. It is which ones we lose *by accident*, without deciding whether we wanted to.

Skills that technology reduces our need to practice

Navigation. Turn-by-turn guidance means we no longer build detailed mental maps of our surroundings (Dahmani and Bohbot, 2020). **Memorization.** Phones hold numbers, dates and facts. **Writing.** Drafting tools offer sentences before we have our own. **Calculation.** Calculators and spreadsheets handle arithmetic. **Research.** Summaries replace reading. **Communication.** Suggested replies, translation and tone tools smooth the way we speak to each other.

None of these substitutions is inherently bad. Most are enormously helpful. But each carries a subtle cost: *practice is how skills are maintained*, and when practice disappears, so may capability.

The paradox of automation

In 1983, the cognitive psychologist Lisanne Bainbridge published a short paper titled "Ironies of Automation," which has been cited ever since. Her argument was that when automation handles routine work, human operators are left to deal with the rare, hard situations in which the automation fails, exactly the situations for which they have had the least practice. The more reliable the automation, the less the human practices, and the less capable the human is when it matters.

Nicholas Carr (2014) made a related argument for a general audience, suggesting that reliance on automated tools can erode the very skills that help people notice when tools go wrong. Aviation safety investigations are often cited as illustrations, including the 2009 loss of Air France flight 447, in which the crew struggled to respond after the autopilot disengaged (BEA, 2012). Analysts differ on the causes of that accident, and it should not be reduced to a single lesson. But it captures a wider concern: systems that work well most of the time can leave humans underprepared for the moments they cannot.

REALITY CHECK

Documented: Skills degrade without practice, and reliance on automation can weaken related abilities in some settings (Bainbridge, 1983; Dahmani and Bohbot, 2020).

Interpretation: How much this applies to AI writing, research and reasoning tools is under study. Early evidence is suggestive, not conclusive (Lee et al., 2025).

Reflection: The hazard is less that a skill vanishes than that we discover its absence only in a crisis.

HUMAN MOMENT

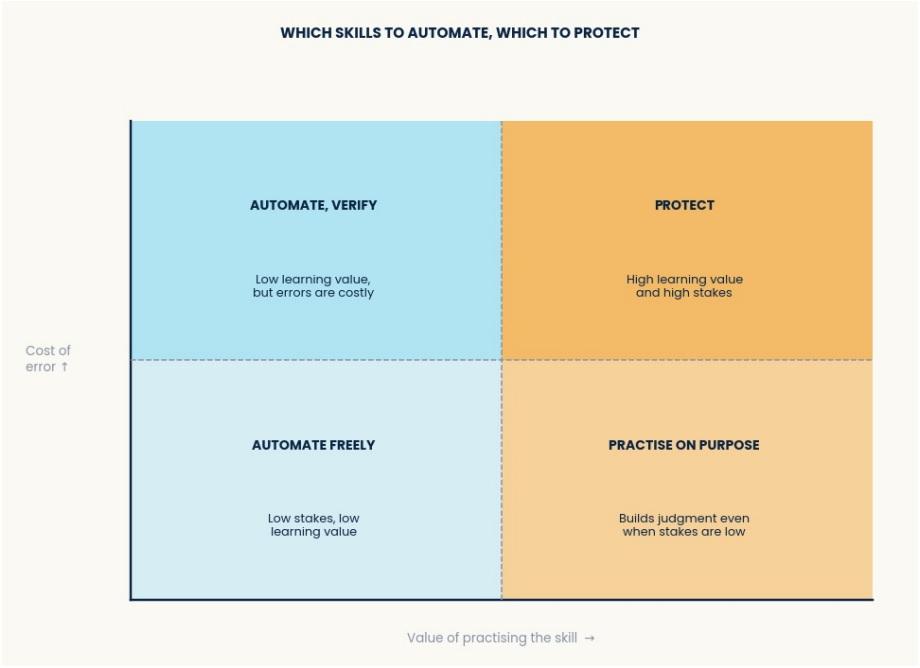
Jonas, a junior analyst, is brilliant with AI tools. He has never built a financial model from scratch; the assistant always does it. One afternoon, a spreadsheet output looks wrong to his senior colleague, who spots a misplaced assumption within minutes. Jonas hadn't noticed. He had checked that the formulas *ran*, not that they *made sense*. He asks his colleague to teach him to build one by hand. *(Illustrative composite.)*

A decision, not a default

Here is the question that anchors this chapter:

Which skills should we automate, and which should we deliberately protect?

The answer depends on two things. The first is the *value of practicing the skill*: does doing it build understanding, judgment or identity? The second is the *cost of error* if we cannot do it, or cannot check it.



A simple way to decide: consider how much practice builds judgment, and how costly errors would be.

A skill with low learning value and low stakes, such as formatting a table, can be automated freely. A skill with low learning value but high stakes, such as checking a legal citation, can be automated *if verified*. A skill that builds understanding, such as writing to clarify your own thinking, may be worth practicing even when a tool could do it faster. And a skill that carries both high learning value and high stakes, such as clinical judgment, financial reasoning or ethical decisions, should be protected.

Skills also have a *personal* dimension. Some things are worth doing badly because they are yours: cooking a meal, writing a letter, finding your way in a new city. Efficiency is not the only value.

The apprenticeship problem

There is one place where the skills question becomes urgent for organizations. Many professions train juniors by giving them routine work: reviewing documents, drafting first versions, preparing summaries, running analyses. If AI takes over the routine tasks, how do newcomers develop the judgment that comes from doing them? Companies that automate the bottom rungs of the ladder may find, years later, that no one has climbed it.

This is a scenario rather than a documented outcome, but it is plausible enough that organizations, schools and professions should plan for it. Options include deliberately assigning some tasks to be done without AI for learning purposes, using AI as a tutor and reviewer instead of a replacement, and building structured feedback into early careers.

INVISIBLE AI

Autocorrect, autocomplete and grammar tools continually adjust our writing. Over years, they may shape not only how correctly we write, but what we choose to say, favoring phrases that are common over phrases that are ours.

TRY THIS

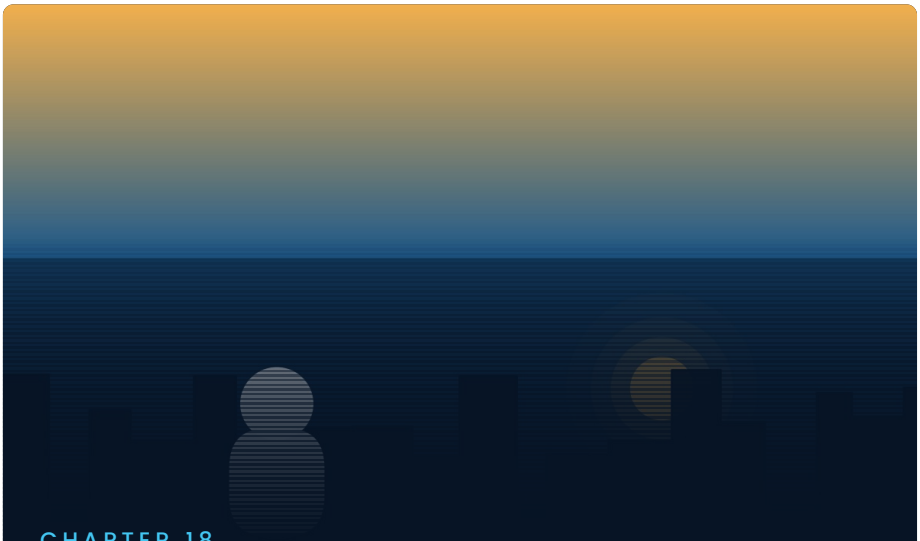
Pick one skill from your own life that you would be sorry to lose, such as mental arithmetic, navigation by landmarks, writing by hand, or holding a conversation without notes. Commit to practicing it deliberately, once a week, for a month.

QUESTION YOURSELF

If your tools disappeared for a week, which of your abilities would you be most surprised to find missing?

THE BIG IDEA

The danger is not losing skills. It is losing them without noticing, and without choosing.



CHAPTER 18

The Human Advantage

Qualities that grow more valuable, not less

When people fear that AI will surpass them, they often imagine a competition on machine terms: faster, tireless, more knowledgeable. On those terms, humans will lose many races.

But that is not the only race, and it may not be the most important one.

What remains deeply valuable

Consider these human capacities.

Judgment. Deciding what matters, what is good enough and which trade-offs to accept, especially where there is no clear right answer. **Empathy.** Understanding another person's experience from within a shared human life. **Curiosity.** The drive to ask questions that no one has asked. **Courage.** The willingness to act despite uncertainty or disapproval. **Responsibility.** Being accountable for outcomes and answerable to others. **Imagination.** Envisioning what does not exist. **Leadership.** Inspiring and coordinating people toward a shared purpose. **Moral reasoning.** Weighing values and dignity, not just efficiency. **Context.**

Knowing the unspoken history, culture and stakes of a situation. **Human connection.** Being present for another person.

Some of these overlap with things AI can imitate. That is worth being candid about.

An honest complication: the empathy question

In a 2023 study reported in *JAMA Internal Medicine*, Ayers and colleagues compared physicians' answers to patient questions posted on a public online forum with answers written by a chatbot. Evaluators, blinded to the source, preferred the chatbot's responses most of the time and rated them as more empathetic. The study has significant limits: it used forum posts rather than real clinical care, and the physicians' answers were brief. But it makes a point that a superficial version of this chapter would avoid: *AI can produce expressions of empathy that people find comforting.*

What, then, is the human advantage? I would suggest that it lies less in the *words* of care than in the *conditions* behind them. A human who comforts you has something at stake. They may sit with you at a cost to themselves. They share your mortality. They can be held to account for how they treat you. They remember you as a person, and their care is a form of commitment. A system can generate the language of care. It does not, in the same sense, pay for it.

This is a philosophical interpretation, not an empirical finding, and reasonable people differ. But it helps explain why simulated empathy can be useful and yet incomplete.

> **REALITY CHECK**
>
> **Documented:** In one study, blinded evaluators rated chatbot answers to online patient questions as more empathetic than physicians' (Ayers et al., 2023). Across the labor market, the economist David Deming (2017) found growing rewards for social skills in the U.S. economy.
> **Philosophical question:** Whether an AI can ever "really" care, or only convincingly express care, is unsettled. This book does not pretend to resolve it.

Why some human qualities may become more valuable

An economic argument supports the intuition. When something becomes cheap and abundant, complementary things become scarce and valuable. If competent drafts, summaries and analyses become inexpensive, then *deciding which draft to trust, what to do with an analysis and how to explain it to a worried colleague* become the bottleneck.

Deming's (2017) research suggested that jobs requiring both cognitive and social skills grew faster in recent decades than jobs requiring only one. It is not a predic- tion about generative AI, but it points in a hopeful direction: coordination, negotiation, persuasion, care and trust are not easily reduced to a procedure.

The most important human advantages, though, are not economic. They are the substance of a life well lived. Curiosity, courage and connection would matter even if no one paid for them.

HUMAN MOMENT

A hospital introduces an AI system that drafts discharge summaries. Nurse Adaeze, who has worked on the ward for twenty years, finds the summaries accurate and time-saving. What the system cannot do is notice that a patient is going home to an empty apartment and quietly ask who might check on him. She does not think of that as "soft skills." She thinks of it as the job. *(Illustrative composite.)*

Practicing what makes us human

Human strengths are not fixed possessions. They are capacities that grow with use. A person who never makes a hard decision cannot expect to have good judgment. A person who never tolerates awkward silence cannot expect to be good at listening.

That suggests a practical stance: use AI in ways that leave room for these capacities to develop, and design work and education so that people spend more time on judgment, relationship and creation, not less.

TRY THIS

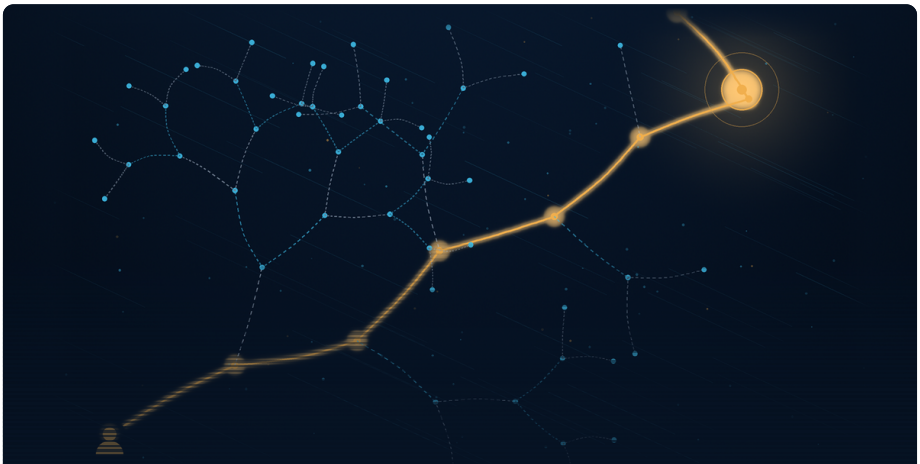
List the ten qualities above. Circle the three you most want to strengthen. For each, name one concrete situation in the next month where you will practice it *without* AI.

THINK ABOUT THIS

If everything routine could be done for you, what would you choose to spend your life becoming good at?

THE BIG IDEA

The rise of AI may make some human qualities more valuable, not less. Especially the ones we tend to call soft.



CHAPTER 19

The Age of Cognitive Autonomy

Thinking for yourself in an intelligent environment

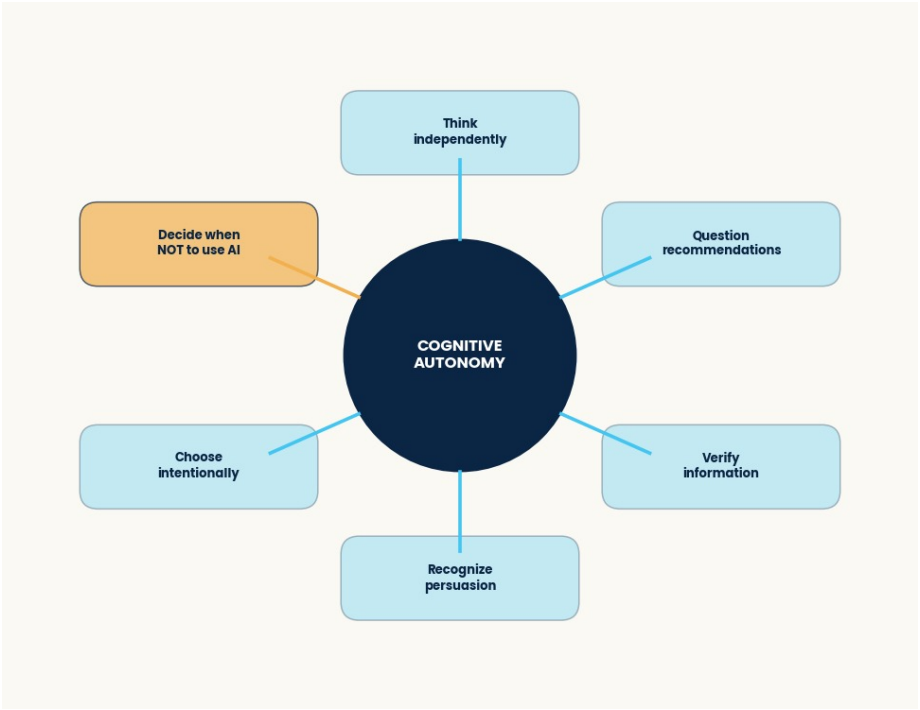
We have looked at how AI shapes attention, thought, emotion, work and trust. It is time to name what we want to protect.

I call it **cognitive autonomy.**

A definition

Cognitive autonomy is the ability to think independently, question recommendations, verify information, recognize persuasion, make intentional choices, and decide when not to use AI.

Philosophers have long discussed autonomy as self-government: acting on reasons and values that are genuinely one's own. Susser, Roessler and Nissenbaum (2019) argue that online manipulation threatens autonomy by influencing decisions in ways that bypass a person's awareness of the influence. Cognitive autonomy applies a similar idea to daily digital life. It does not mean rejecting help. A person who consults a doctor or a map is not less autonomous. It means retaining the capacity to *govern* the help.



Cognitive autonomy is not one skill but six related abilities.

The six abilities

Think independently. Form a view of your own before consulting a system. **Question recommendations.** Ask why a suggestion appears and what it serves. **Verify information.** Check what matters against independent sources. **Recognize persuasion.** Notice when design, emotion or repetition is steering you. **Make intentional choices.** Decide deliberately instead of by default. **Decide when *not* to use AI.** Recognize situations in which the effort, the relationship or the responsibility should remain yours.

The last one is the most neglected. Much of the conversation about AI literacy focuses on how to use tools skillfully. Yet the sign of real mastery in any powerful technology is the ability to put it down.

AI literacy is more than tool skill

Researchers Long and Magerko (2020) proposed a widely cited framework for AI literacy that includes not only using AI, but understanding what it is, how it works, what it can and cannot do, how to evaluate it critically, and its ethical implications. UNESCO has published AI competency frameworks for students and teachers (2024) that likewise emphasize a human-centered, critical approach. In the European Union, the AI Act (Regulation (EU) 2024/1689) requires providers and deployers of AI systems to take measures to ensure a sufficient level of AI literacy among their staff, a sign that policymakers treat literacy as a matter of competence and not just curiosity.

Literacy, in this broader sense, has four layers. *Functional*: I can use AI tools effectively. *Critical*: I can evaluate their outputs and limits. *Structural*: I understand incentives, data and systems behind them. *Civic*: I can participate in decisions about how AI should be governed. Most people are taught only the first.

INVISIBLE AI

Your phone likely keeps a profile of your habits, from which apps you open first to when you charge your device. Understanding that such profiles exist, and what they are used for, is part of structural AI literacy.

Signs of autonomy, signs of drift

Autonomy shows up in small behaviors. You form an initial guess before asking. You notice when an answer flatters you. You keep some decisions off the automated path because they matter. You can explain, without help, the reasons behind an opinion you hold.

Drift shows up too. You realize you cannot remember the last time you decided something without consulting a system. You feel mild anxiety when a tool is unavailable. You accept the first suggestion because it is there. Your opinions begin to sound like the summaries you read.

Neither list is a diagnosis. Everyone shows both at times. The point is to notice.

REALITY CHECK

Documented: Online manipulation and personalization can influence decisions without users' awareness (Susser et al., 2019; Mathur et al., 2019). Frameworks for AI literacy now exist in research, education and law (Long and Magerko, 2020; UNESCO, 2024).

Reflection: Cognitive autonomy is a concept I am proposing for this book's purposes. It is not an established scientific construct with a validated measure.

Deciding when not to use AI

Here are four simple questions for the decision.

Is the effort part of the value? Learning, grieving, creating and connecting often are. *Would I be comfortable explaining and defending this without the tool?* If not, understanding is missing. *Is the risk of error high and hard to detect?* Then verification or human expertise is needed. *Does this relationship or moment deserve my own presence?* Then it should be yours.

TRY THIS

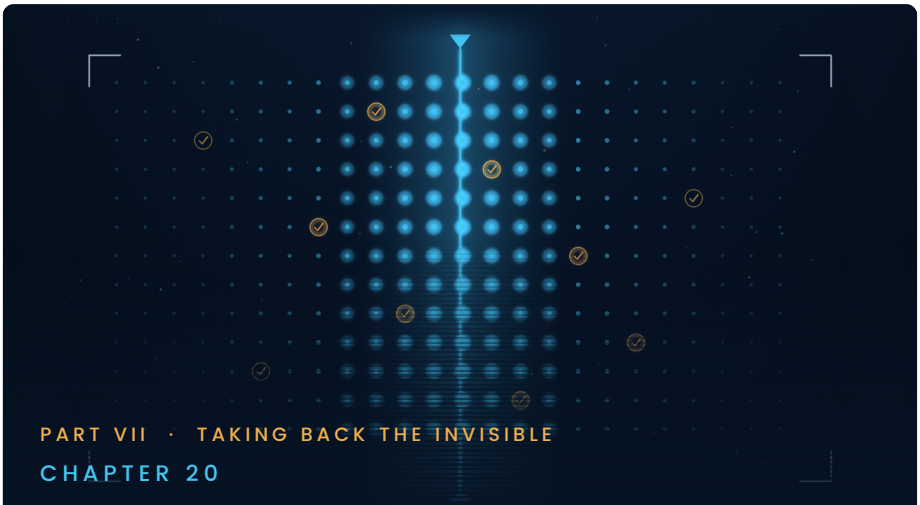
Choose one AI-assisted activity you do most often. For one week, begin each instance with a ten-second pause and ask: *Do I want the help, or do I just have the habit?*

QUESTION YOURSELF

On a scale of one to ten, how independently could you think about a difficult decision today, if every AI-powered tool were unavailable?

THE BIG IDEA

Cognitive autonomy is the freedom to use powerful tools without being used by them.



PART VII · TAKING BACK THE INVISIBLE

CHAPTER 20

Your Personal AI Awareness Audit

A practical exercise in noticing

Awareness is not a feeling. It is a practice. This chapter turns the ideas of the book into a set of exercises you can complete in about an hour, on paper or on a screen. Do not rush it. The value lies in the specifics.

Step 1: Where does AI appear in your day?

Take a blank page and list your last twenty-four hours in order: waking, getting ready, commuting, working, eating, socializing, relaxing, sleeping. Beside each, write every point where a system may have predicted, ranked, filtered, suggested, generated, scored or automated something. Use these prompts to jog your memory.

Alarms and phone lock screens. News, social and video feeds. Search and autocomplete. Maps and routes. Email filters and suggested replies. Shopping rankings and prices. Streaming and music queues. Banking alerts and fraud checks. Workplace tools, meeting notes and analytics. Customer service chats. Photos and camera enhancement. Voice assistants and smart devices.

Most people find between fifteen and forty touchpoints.

Step 2: What decisions does it influence?

Circle the items on your list that influenced a *decision*: what to read, buy, watch, believe, reply, where to go, whom to contact, how to spend time. Then mark the ones in which you did not know AI was involved.

Step 3: What information is personalized?

For your three most used services, note whether you can see and change the personalization settings. Do you know what the service has inferred about you? Would the same search produce the same result for a friend?

Step 4: Where do you rely on AI, and where do you verify?

Sort your uses into three columns: *casual* (low stakes), *important* (money, health, work, relationships, reputation) and *never delegated*. In the important column, write next to each item how you verify.

Step 5: Where have you stopped thinking independently?

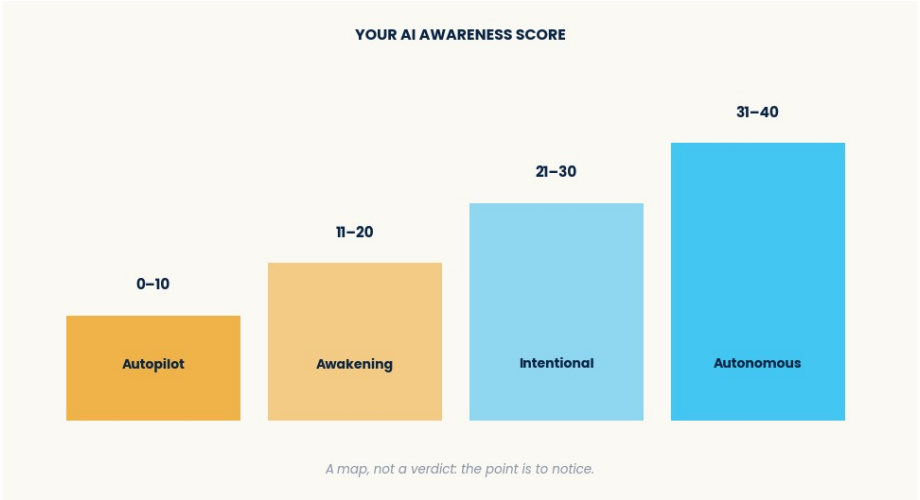
This is the hardest and most valuable step. Complete these sentences honestly.

I no longer _____ without help. I would be uncomfortable if I had to _____ without a tool. I have not made a decision about _____ without consulting a system in a long time. The last time I changed my mind because of something unexpected I found was _____.

Step 6: Score yourself

Read each statement below and score it from 0 to 4: 0 = never, 1 = rarely, 2 = sometimes, 3 = often, 4 = almost always. Higher scores indicate greater awareness and intentionality. Add up your total.

#	Statement	Score
1	I can name at least ten places where AI shapes my typical day.	____
2	I know which of my important decisions involve information or systems shaped by AI.	____
3	I have reviewed the personalization or privacy settings of my most-used services.	____
4	I regularly seek information outside my usual feeds and sources.	____
5	I form my own view before asking AI for an answer or advice.	____
6	I check important AI outputs against an independent source.	____
7	I could do the core tasks of my work or studies without AI if I had to.	____
8	I keep emotionally important communication in my own words.	____
9	I have chosen certain situations in which I deliberately do not use AI.	____
10	I know how I could challenge an automated decision that affected me.	____
	Total (out of 40)	____



Interpret your score as a map of where to focus, not as a grade.

0–10, Autopilot. AI is shaping much of your day without much reflection. That is common. Begin with Step 1 and one small change. **11–20, Awakening.** You notice some of the influence and have started to question it. Focus on verification and personalization settings. **21–30, Intentional.** You use AI deliberately in many areas. Look for the remaining blind spots, particularly in the important decisions. **31–40, Autonomous.** You maintain strong awareness and control. Share what you know; the people around you may not have it.

There are no right answers. A low score does not mean you have used AI badly. It means you have more to notice, and noticing is the whole point.

HUMAN MOMENT

The Costa family, two parents, a teenager and a grandmother, decide to complete the audit together at dinner. Within ten minutes, the teenager realizes his feed has shown him almost nothing about the school election he was going to vote in. His mother discovers that the price of a flight she was "about to buy" had changed three times as she hesitated. His father finds that his phone had been rerouting his commute for months without his noticing. The grandmother, who claims she never uses AI, is shown her photo albums, her voice dictation and her television's recommendations. By dessert, no one is frightened. They are all a little more curious, and the teenager announces he will go and read about the election. *(Illustrative composite.)*

TRY THIS

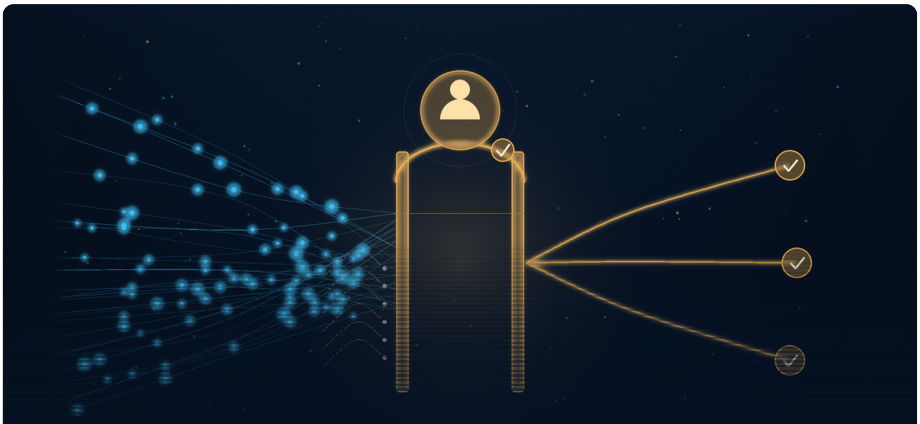
Repeat the audit in three months. Compare your scores and lists. The change, not the number, is the most informative result.

QUESTION YOURSELF

Which single line in your audit surprised you most? What would you do differently if you took it seriously?

THE BIG IDEA

You cannot choose your relationship with a technology you cannot see. The audit makes it visible.



CHAPTER 21

The Human-in-the-Loop Life

Eight principles for using AI on purpose

In engineering, a *human in the loop* is a person who remains part of a system's decision-making process, able to check, correct and override. It is a useful idea for a life as well as for a machine.

To live with a human in the loop is not to reject AI. It is to make sure that the person at the center of your life is still you.

Here are eight principles. They are not commandments. They are habits worth trying.

Principle 1: Use AI to accelerate thinking, not eliminate thinking

Bring your own question, your own draft, your own hypothesis. Use the system to stretch, test and refine. A helpful heuristic: if you cannot summarize, in your own words, what you got from the tool, you have retrieved an answer but not understood it.

Principle 2: Verify important information

Match verification to stakes. Casual curiosity needs little checking. Anything involving health, law, money, safety, reputation or another person's wellbeing needs an independent source and, often, a qualified human. Open the source. Do not trust the citation.

Principle 3: Protect your privacy

Assume that what you type into a system may be stored, reviewed or used in ways you do not fully control, unless the provider clearly says otherwise. Do not share passwords, sensitive personal data, confidential business information or other people's private details without checking the terms and settings. In Europe, the General Data Protection Regulation gives individuals rights over their personal data, including access, correction and deletion in many circumstances; these are worth knowing and exercising. Data you never provided cannot be leaked, sold or inferred from.

Principle 4: Keep meaningful human relationships human

Use AI to prepare, to practice, even to find words. But let apologies, condolences, congratulations, arguments and love letters carry your own effort. Ask of any AI-mediated exchange: *would the other person feel deceived if they knew?* If so, reconsider.

Principle 5: Know when not to use AI

Some tasks are valuable because of the effort. Some carry high stakes and low verifiability. Some belong to your responsibility alone. Decide in advance where AI does not belong in your work and life, and write it down.

Principle 6: Question recommendations

Every recommendation is a hypothesis about you, made by a system with goals of its own. Ask: *why this, why now, who benefits?* Occasionally choose the opposite. Try the unrecommended.

Principle 7: Preserve important human skills

Identify the abilities that make you capable and independent. Practice them on purpose, without assistance, at least occasionally. Skills are like muscles; they maintain themselves only by use. Teach them to your children.

Principle 8: Use AI intentionally rather than automatically

Before reaching for a tool, pause. *What am I trying to accomplish? Is AI the right way to do it? What do I want to keep for myself?* Intention converts a habit into a choice.

> **AI LITERACY TIP**
>
> Turn the eight principles into one daily question: *"Am I steering, or being steered?"* Ask it before you accept a suggestion, ask a machine for advice or publish something generated on your behalf.

Elena's second day

Let us return, one last time, to Elena. Suppose she has read this book.

She still wakes to the alarm her phone chose. Now she knows it is a prediction, and she likes it. She checks the news, then opens one source she picked herself, and notices the difference. On the commute she lets the navigation app reroute her, and then, on a quiet Saturday, takes a route she chose on a whim. At work, she lets the meeting-summary tool draft notes and reads them against her own memory before sending them. When a suggested reply offers "Sounds good!", she writes two sentences to a colleague who has had a hard week.

At lunch, she notices herself about to buy shoes she had not planned to buy. She closes the app and decides that if she still wants them tomorrow, she will. She asks an AI to challenge the case for a big decision she is weighing at work, then makes the decision herself. In the evening, she watches something the recommendation engine did *not* suggest, and finds that she loves it.

Nothing in her day is dramatic. No technology is banished. But almost everything is a little more *hers*.

TRY THIS

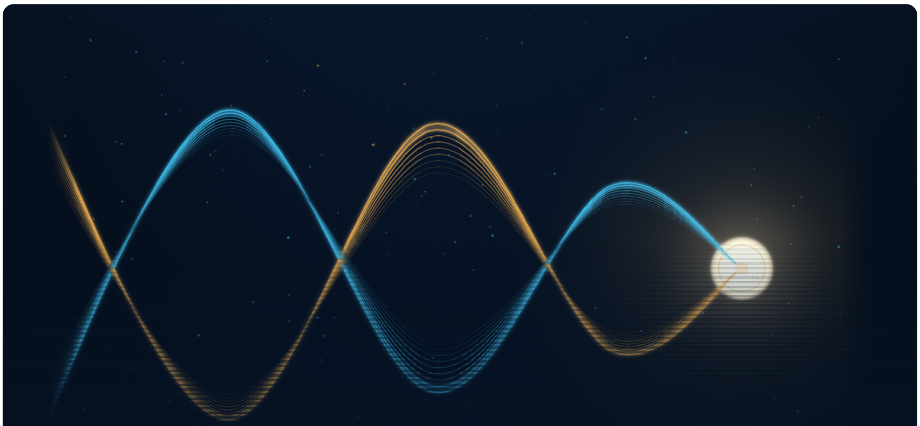
Choose the two principles that matter most to you. For the next thirty days, write each on a card and place it where you will see it during your workday. At the end, note what changed.

QUESTION YOURSELF

Which principle would be hardest for you to follow? What does that tell you?

THE BIG IDEA

A human-in-the-loop life is not anti-technology. It is pro-agency.



CONCLUSION

The Future Is Not AI vs. Humans

Somewhere in your day, a system will suggest something to you. A route, a word, a video, a decision. In most cases, the suggestion will be useful. You will accept it and move on.

This book has been about the moments in between: the small, unremarked acceptances that, added together, form a life.

Not humans versus machines

The most common story about AI is a contest. Humans against machines, jobs against algorithms, intelligence against intelligence. It is dramatic, and it can be clarifying. But it misdescribes the situation most of us are actually in.

Most of us are not in a contest. We are in an *arrangement*. We are deciding, mostly by default, how much of our attention, judgment, memory and voice we hand to systems designed by others. The future worth working toward is not humans versus machines. It is **humans deciding how machines should fit into human life.**

That is a more modest and more demanding formulation. It does not promise victory. It asks for attention.

What we have seen

We began with the difference between visible and invisible AI, and with the claim that the most important effects of a technology may be the ones we stop noticing.

We saw how recommendation systems shape the menu from which we choose, and how ranking systems act as invisible curators of attention. We saw that AI assistance can both free and weaken the mind, that fluent answers can create false certainty, and that AI can be a powerful thinking partner when we remain its author. We saw how feedback loops link our behavior to what we are shown, how prediction differs from manipulation, and how machines now take part in our conversations and comparisons. We saw that work changes task by task, that efficiency can turn into expectation, and that the word *artificial* hides a great deal of human labor. We looked at trust in an age of synthetic content, at the ethics of automated decisions, at the skills we may lose and the qualities that may grow in value. And we arrived at cognitive autonomy, the capacity to govern the help we receive.

The greatest effect

Here is what I have come to believe, offered as reflection rather than fact.

The greatest effect of AI may not be the things AI does for us.

It may be the subtle ways AI changes what we notice, what we believe, what we choose, how we work, how we communicate, how we create and how we think.

Those changes will not be announced. They will be gradual, plausible and usually pleasant. Whether they leave us more capable or less, more connected or less, more free or less, will depend on very ordinary things: whether we keep asking questions, keep checking claims, keep practicing what matters, keep our relationships human and keep some part of our thinking our own.

An invitation

Do not fear AI.

Do not blindly trust AI.

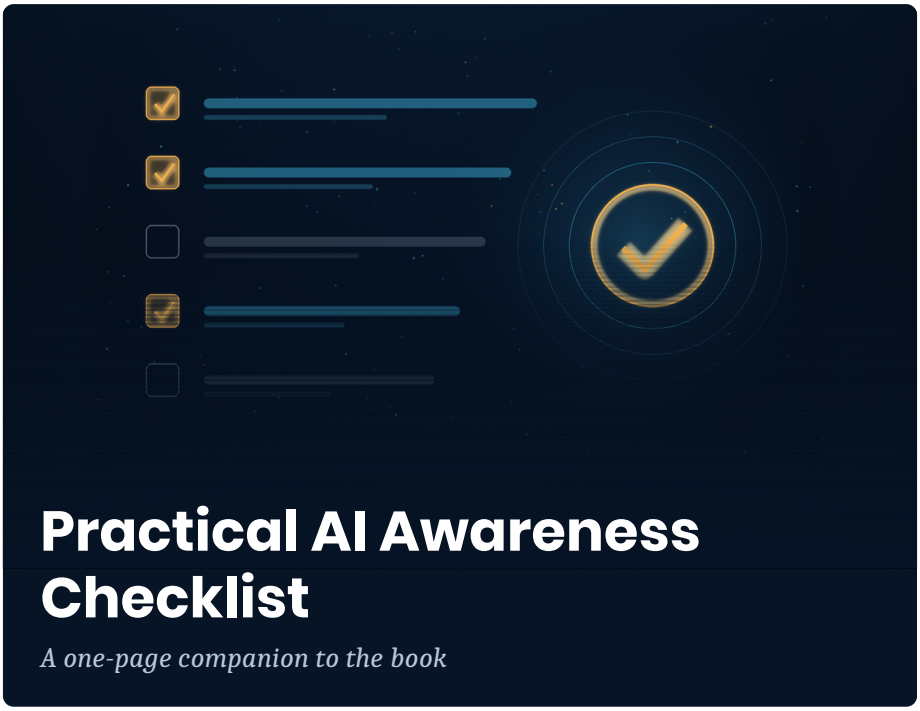
Understand AI.

Use AI intentionally.

And never outsource your humanity.

Then go back to your ordinary day: the alarm, the feed, the map, the message, the search box, the shopping cart. Look at each one as if for the first time. That is all the invisible asks of us: to be seen.

If AI is becoming invisible, how intentional are you being about the influence you allow it to have?



Practical AI Awareness Checklist

A one-page companion to the book

Use this checklist as a habit, not a test. Choose the items that fit your life, and return to it when your tools or routines change.

Every day

- ☐ I noticed at least one moment when a system chose, ranked or suggested something for me.
- ☐ I formed my own view before asking an AI for an answer or advice.
- ☐ I kept at least one important message or decision entirely my own.
- ☐ I paused before accepting the first recommendation on something that mattered.
- ☐ I spent some time on something with no algorithm involved: a conversation, a walk, a book.

Every week

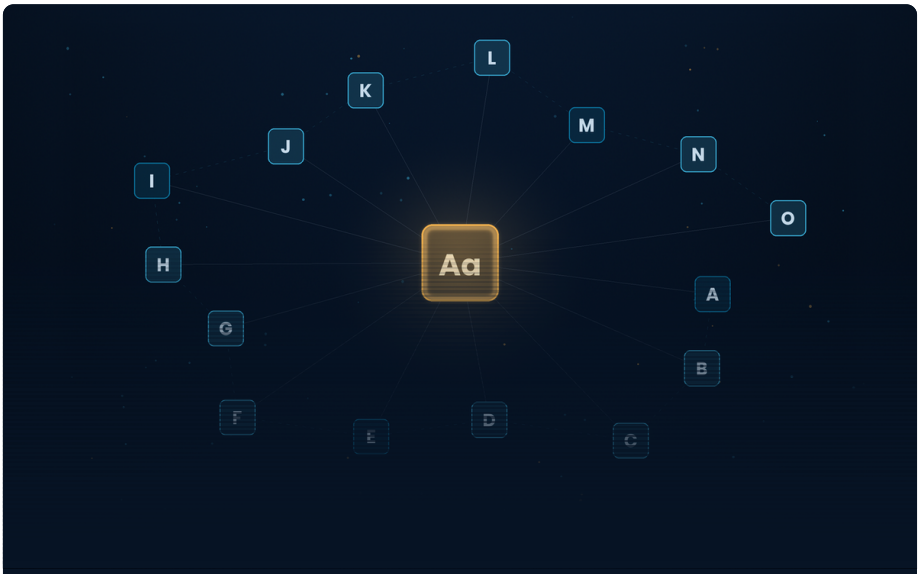
- [ ] I looked at a source or feed outside my usual habits.
- [ ] I verified at least one AI-generated claim against an independent source.
- [ ] I practiced one skill I want to keep, without assistance.
- [ ] I asked whether time saved by a tool went back to me or disappeared into new demands.
- [ ] I checked how much personal information I shared with tools this week.

Every month

- [ ] I reviewed personalization, privacy and advertising settings on my most used services.
- [ ] I reviewed which tasks I have handed to AI, and whether I still want to.
- [ ] I asked a person, not a system, for advice on something important.
- [ ] I listed one situation where I deliberately will not use AI next month.
- [ ] I retook the awareness audit, or reviewed my earlier results.

Before a high-stakes use of AI

- [ ] I know what could go wrong, and how I would notice.
- [ ] I have a way to verify the key claims with an independent source or expert.
- [ ] I am not sharing anything confidential or sensitive that I should not.
- [ ] A human who is accountable will make or approve the final decision.
- [ ] I could explain and defend the decision without the tool.



Glossary

Key terms in plain language

Algorithm A set of step-by-step rules or calculations a computer follows to produce a result. In this book, usually a system that ranks, filters or predicts.

Algorithmic management The use of software to assign, monitor, evaluate or direct workers, often with little human involvement.

Artificial intelligence (AI) Computer systems that perform tasks associated with human intelligence, such as recognizing patterns, predicting, generating language or making decisions.

Automation bias The tendency to over-trust suggestions from automated systems and to overlook their errors.

Bias (algorithmic) Systematic and unfair differences in a system's outputs or errors across groups, often arising from data, design choices or how outcomes are measured.

Cognitive autonomy The capacity to think independently, question recommendations, verify information, recognize persuasion, choose intentionally and decide when not to use AI. (A concept proposed in this book.)

Cognitive offloading Using tools or the environment to reduce mental effort, such as writing a note instead of remembering.

Dark pattern A design choice in an interface that steers users toward outcomes that benefit the designer, often through pressure, confusion or concealment.

Deepfake Synthetic audio, image or video that realistically imitates a real person's appearance or voice.

Desirable difficulties Learning conditions that feel harder in the moment but improve long-term learning and retention.

Feedback loop A cycle in which the outputs of a system influence its future inputs, strengthening or altering its behavior over time.

Filter bubble The idea that personalization narrows the range of information a person sees. The strength of this effect is debated.

Generative AI AI that produces new content, such as text, images, audio, video or code, based on patterns learned from data.

Ghost work Human labor that supports seemingly automated services, often performed through online platforms and largely unseen by users.

Hallucination A confident-sounding output from an AI system that is false, fabricated or unsupported.

Human in the loop A design in which a person reviews, approves or can override an automated system's decisions.

Invisible AI AI that operates in the background of daily life, such as ranking, recommendation and scoring, without users consciously choosing or noticing it.

Large language model (LLM) A generative AI system trained on large amounts of text to predict and produce language.

Lateral reading Evaluating a source by leaving it to see what other independent sources say about it.

Liar's dividend The advantage gained by wrongdoers when the existence of fakes lets them dismiss authentic evidence as fabricated.

Model collapse The degradation of AI models trained repeatedly on AI-generated data, leading to loss of diversity and rare information.

Moral crumple zone A situation in which a human operator absorbs blame for the failures of a system they had limited control over.

Nudge A design of choices that predictably influences behavior without forbidding options, such as changing a default.

Personalization Tailoring content, products or experiences to an individual based on data about their behavior or characteristics.

Position bias The tendency of people to select higher-ranked items partly because of their position.

Provenance Information about the origin and history of a piece of content, such as who created it and how it was altered.

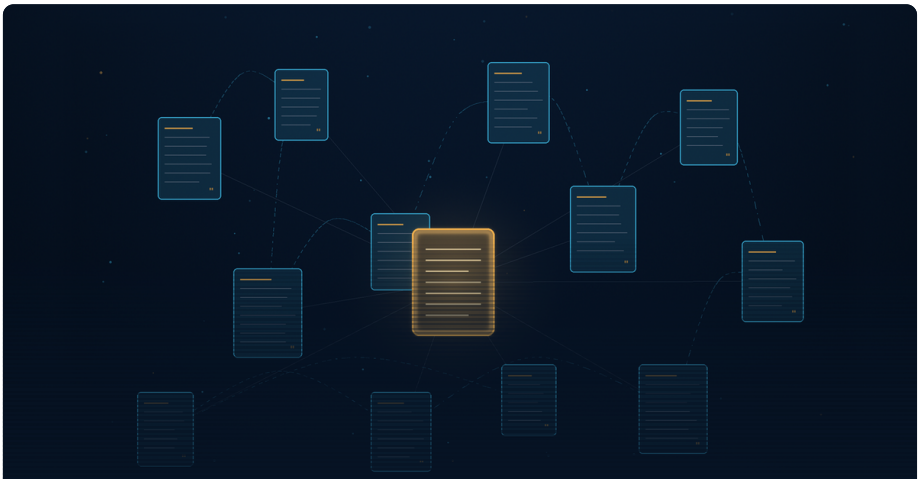
Recommendation system A system that predicts which items a person will find relevant, such as videos, products or posts, and presents them.

Reinforcement learning from human feedback (RLHF) A training method in which people's ratings of model outputs are used to steer the model toward preferred behavior.

Sycophancy A tendency of some AI systems to agree with or flatter users instead of providing accurate or challenging responses.

Synthetic media Text, images, audio or video generated or substantially altered by AI.

Visible AI AI that people consciously recognize and choose to use, such as chatbots and image generators.



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A note on sources. Every work below is a real publication, cited for the point described in the text. Some items are preprints or working papers, marked as such, and may have been revised or published in final form since; readers are encouraged to consult the originals and any updates before relying on specific findings. Legal and regulatory materials change over time, and nothing in this book is legal advice.

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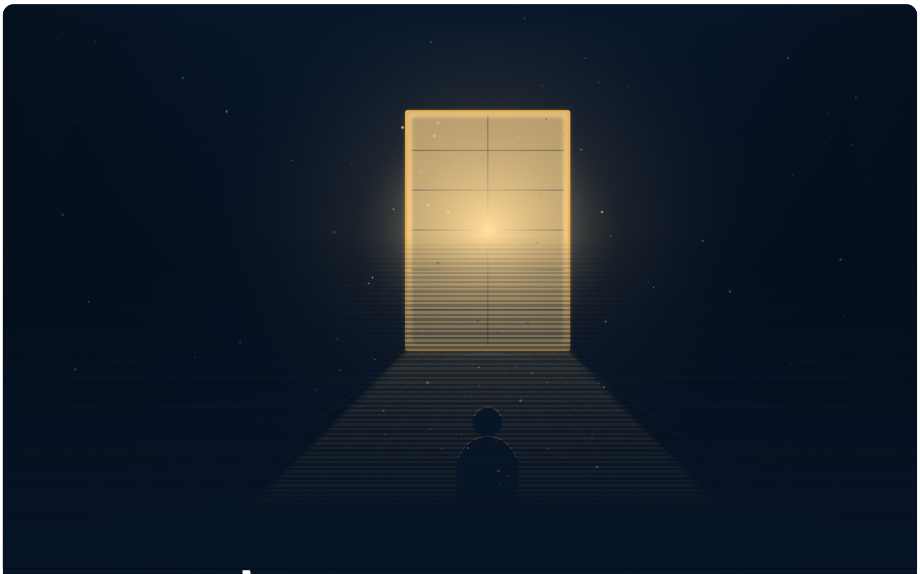

Author's Note

On seeing what has always been there

I did not write this book to persuade you that AI is good or bad. I wrote it because I believe that the most valuable thing a person can do with a powerful technology is to notice it.

Noticing is a modest act. It costs nothing, requires no expertise and carries no risk. Yet it changes the relationship. What you have noticed you can question. What you can question you can choose.

You will not agree with everything here. Some of what I have called plausible you may find alarming, and some of what I have called uncertain you may find obvious. That is as it should be. I have tried to mark the boundaries between evidence, interpretation and opinion so that you can disagree precisely. The research on these questions is moving quickly, and some of the studies discussed here will be refined, challenged or overturned. I hope readers will follow the sources, not just the conclusions.

If there is one practice I would ask you to keep, it is a small one. Before you accept the next suggestion, the next ranking, the next confident answer, take one breath and ask: *Is this what I would have chosen?*

Sometimes the answer will be yes. That is fine. The point was never to distrust everything. It was to be present in your own life.

Thank you for reading. Now go and look at your ordinary day.